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Chapter 7

Plant Kingdom

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The plant kingdom has been classified in various ways by different scientists. Here, we will describe Plantae under **Algae, Bryophytes, Pteridophytes, Gymnosperms** and **Angiosperms**.

CLASSIFICATION SYSTEMS

There are three main types of systems of classification, *i.e.*, artificial, natural and phylogenetic.

Artificial System of Classification

(a) ***Basis*** **:** The earliest systems of classification were **artificial** and involves usage of one or few morphological characters for grouping of organisms. For example, classification within the angiosperms was based only on morphological characters such as habit, colour, number and shape of leaves or on the androecium structure (system given by Linnaeus).

(b) ***Proponents*** - Aristotle and Linnaeus.

(c) ***Drawbacks*** **:** (i) They separated the closely related species.

(ii) Also equal weightage to vegetative and sexual characteristics were given. Since the vegetative characters are more affected by environment, so this was not acceptable.

Natural System of Classification:

(a) ***Basis*** **:** Organisms are classified on the basis of natural affinities and consider not only external but also internal features like ultrastructure, anatomy, embryology and phytochemistry.

(b) ***Proponents*** **: George Bentham** and **Joseph Dalton Hooker**.

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Phylogenetic System of Classification

(a) ***Basis*** **:** At present phylogenetic classification systems are acceptable which are based on evolutionary relationships between the various organisms, assuming that organisms belonging to the same taxa have common ancestor. The fossils play important role in elucidation of evolutionary relationships.

(b) ***Proponents*** **: Engler and Prantl, Hutchinson, Takhtajan**

We now use information from many other sources too to help resolve difficulties in classification. These become more important when there is no supporting fossil evidence.

BRANCHES OF TAXONOMY

Numerical Taxonomy (Phenetics) : It involves usage of numerical methods for the evaluation of similarities and differences between species with the help of computers.

Steps involved in numerical taxonomy :

1. Numbers and codes are assigned to all the observable characters like plus (+), minus (–), data not available (0).
2. All possible characters are compared by computers by giving equal importance.
3. The organisation and analysis of data forms core of this taxonomy.

   One of the major benefits of this taxonomic method is that hundreds of characters can be considered at the same time.

Cytotaxonomy / karyotaxonomy : It is based on cytological information like chromosome number, structure, behaviour, etc.

Chemotaxonomy : It is based on the chemical constituents of the plant. For example DNA sequence, chemical nature of proteins, crystals (Calcium oxalate or calcium carbonate) and aromatic compounds are used by scientists to resolve confusions in classification.

ALGAE (Father of algology : F. E. Fritsch)

Algae are chlorophyll containing, simple, thalloid (plant body not differentiated into root, stem and leaf) and autotrophic organisms.

The main characteristics of algae are:

1. Algae are largely aquatic, either marine or fresh water. They also occur in habitats like moist stones, soils and wood. Some of them also occur in association with fungi (lichen) and animals (*e.g.*, on sloth bear).
2. Algae are of variable size and forms. The size ranges from the **unicellular** (*Chlamydomonas*), **colonial** (*Volvox*), **filamentous kelps** to massive plant bodies (*Ulothrix* and *Spirogyra*).
3. Vascular tissues are absent.
4. Algae reproduce by vegetative, asexual and sexual methods.
5. **Vegetative reproduction** is by fragmentation. In fragmentation, the parent body breaks into two or more fragments which develops into a thallus.
6. During the **asexual reproduction** different types of spores are produced. Spores are released from the parent body which, on germination give rise to new plants. The most common spores produced are the flagellated (motile) **zoospores**.

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7. Sex organs are non-jacketed and unicellular. Sexual reproduction can be :

(i) **Isogamous :** Fusion of two gametes which are similar in size, either flagellated or non-flagellated is termed as isogamous., *e.g.*, *Ulothrix* (gametes are flagellated and similar in size), *Spirogyra* (gametes are non-flagellated and similar in size).

(ii) **Anisogamous :** Fusion of two gametes which are dissimilar in size is termed as anisogamous. *e.g.*, *Eudorina*.

(iii) **Oogamous :** When a small, motile or non-motile male gamete fuses with a large non-motile female gamete, such type of reproduction is known as oogamous., *e.g.*, *Volvox*, *Fucus*, *Polysiphonia* (both gametes non-motile)

Algae are mainly classified on the basis of their pigments. Flagellation, storage products and chemistry of cell wall are also taken into account. The three classes of algae are **Chlorophyceae** (green algae), **Phaeophyceae** (brown algae) and **Rhodophyceae** (red algae).

Chlorophyceae

1. The pigments, chlorophyll-a, chlorophyll-b, carotene and xanthophyll are present in them. They are usually grass green in colour due to the dominance of pigments, *i.e.*, chlorophyll a and b.
2. These pigments are present in definite chloroplasts. The shape of chloroplast varies in different species. It can be discoid, plate-like, reticulate, cup-shaped, spiral or ribbon-shaped.
3. The plant body may be unicellular, colonial or filamentous.
4. Most of the green algae have one or more storage bodies called **pyrenoids** located in the chloroplasts. Pyrenoids contain **protein** and **starch**. In some food is stored in the form of **oil droplets**.
5. Green algae usually have a rigid cell wall made up of an inner layer of cellulose and an outer layer of pectose.
6. Vegetative reproduction by fragmentation.
7. Asexual reproduction takes place by zoospores produced inside the zoosporangia. Zoospores have 2-8, apically inserted isokont flagella.
8. Sexual reproduction may be isogamous, anisogamous or oogamous.
9. Some commonly found green algae are : *Chlamydomonas*, *Volvox*, *Ulothrix*, *Spirogyra* and *Chara*.

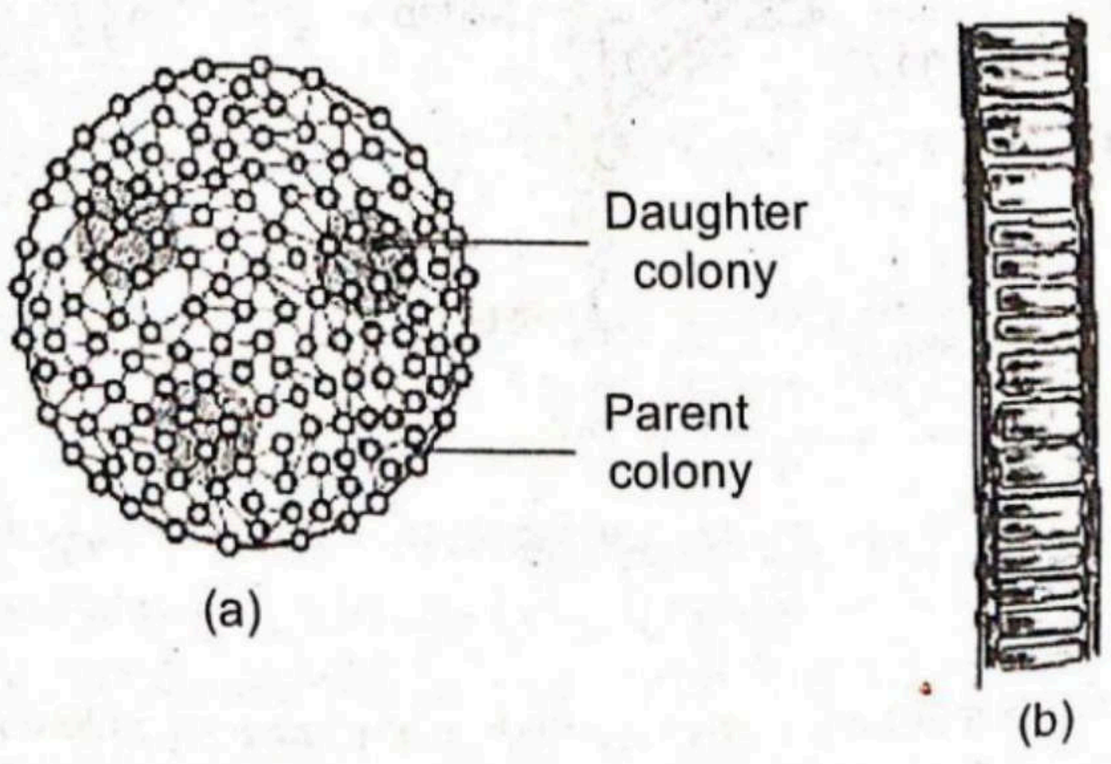


Fig. : Green algae (a) *Volvox*, (b) *Ulothrix*

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Phaeophyceae

1. Brown algae are found primarily in marine habitats.
2. They possess the pigment chlorophyll a and c, carotenoids and xanthophylls.
3. They vary in colour from olive green to various shades of brown depending upon the amount of the xanthophyll called **fucoxanthin**.
4. They show great variation of size and form. In *Ectocarpus* the body is simple branched and filamentous whereas in **kelps** the body is profusely branched *e.g.*, *Laminaria*, *Macrocystis*. Kelps may reach a height of 100 metres.
5. Food is stored as complex carbohydrates which may be in the form of **laminarin** or **mannitol** (alcoholic sugar).
6. Brown algae have cell wall made up of cellulose which is usually covered on the outside by a gelatinous coating of algin. Algin is a hydrocolloid which has good water holding capacity.
7. Their protoplast contains plastids, a centrally located vacuole and nucleus.
8. The plant body is often differentiated into **holdfast** for attachment to a substratum, **stipe** (stalk) and leaf-like photosynthetic organ called **frond**.
9. The large forms often possess **air bladders** for providing **buoyancy**.
10. Vegetative reproduction takes place by fragmentation.
11. Asexual reproduction occurs with the help of biflagellate zoospores that are pear-shaped and have two unequal laterally attached flagella.
12. Sexual reproduction may be **isogamous**, **anisogamous** or **oogamous**. Fusion of gametes takes place in water but in oogamous species, gametes fuse in the **oogonium**. Oogonium is the female reproductive structure, usually a rounded cell or sac containing egg cell or **oosphere** (the non-motile female gamete).
13. The gametes are pyriform *i.e.*, pear-shaped and bear two laterally attached flagella.
14. The common forms of brown algae are *Ectocarpus*, *Dictyota*, *Laminaria*, *Sargassum* and *Fucus*.

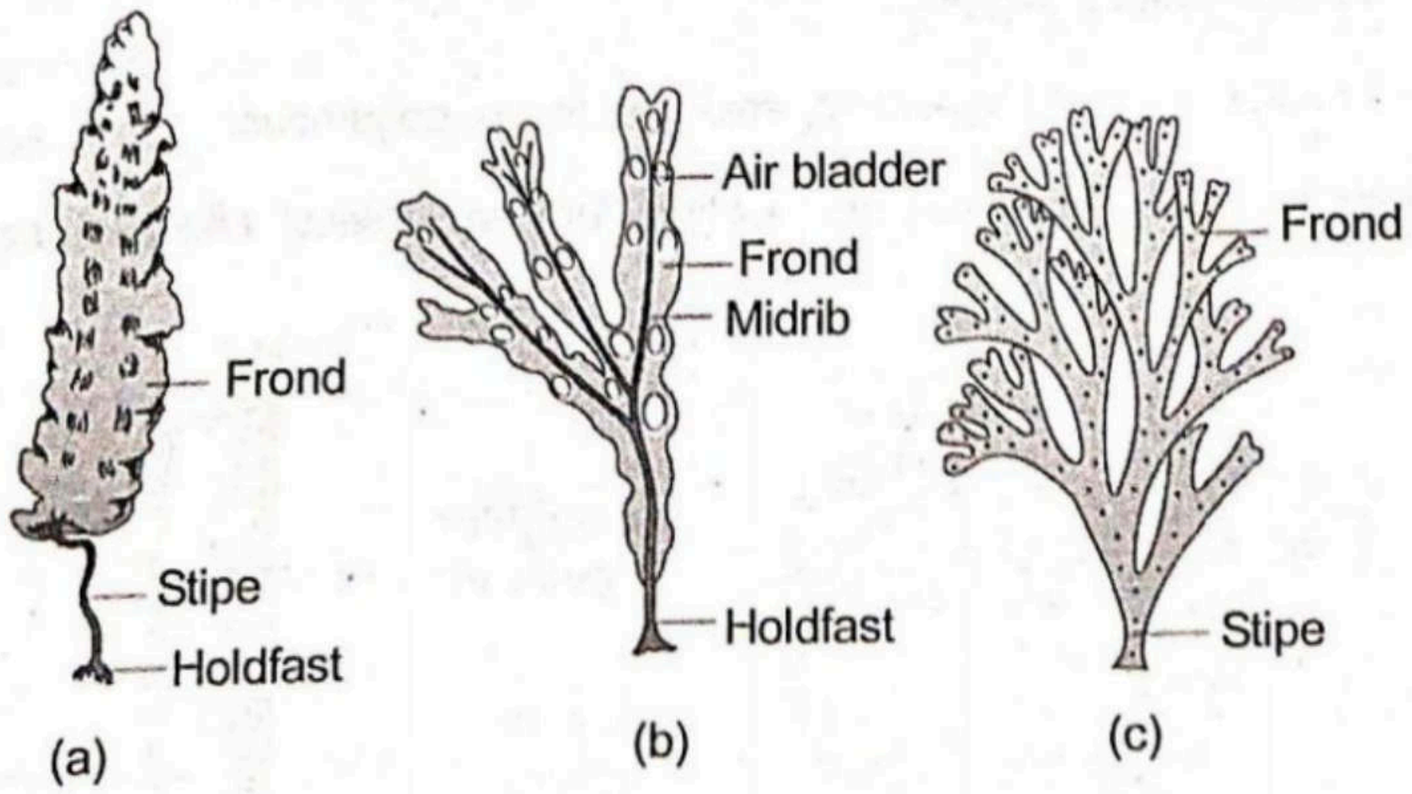


Fig. : Brown algae : **(a)** *Laminaria*, **(b)** *Fucus*, **(c)** *Dictyota*

Rhodophyceae

1. The pigments present in them are r-phycoerythrin, r-phycocyanin, chlorophyll-a and chlorophyll-d. The characteristic red colour is due to the predominance of the red pigment, **r-phycoerythrin**.
2. They are mostly found in marine habitats with abundance in the warmer areas.

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3. They occur in both well-lighted regions close to the surface of water and also at regions where relatively little light penetrates, *i.e.*, at great depths in oceans.
4. Most of the red algae are multicellular. Some of them have complex body organisation.
5. Cell wall is composed of cellulose, pectin and hydrocolloids like **agar, carrageen and funori** (hydro colloids).
6. Food is stored in the form of **floridean starch**. Structurally, it is very similar to amylopectin and glycogen.
7. They reproduce vegetatively by fragmentation.
8. Red algae reproduce asexually by non-motile spores.
9. Sexual reproduction is oogamous and the gametes are non-motile.
10. Complex, post-fertilisation developments occur.
11. The common members are : *Polysiphonia*, *Porphyra*, *Gracilaria* and *Gelidium*.

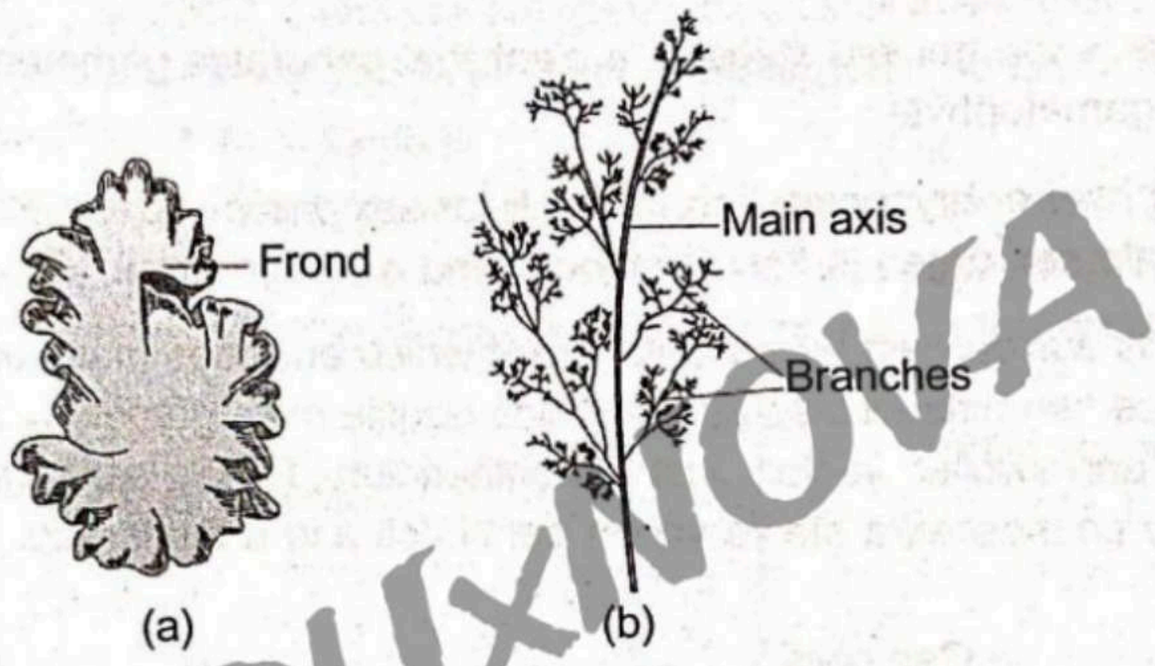


Fig. : Red algae : **(a)** *Porphyra*, **(b)** *Polysiphonia*

Note :

Red algae can reach the maximum depth in sea where no other photosynthetic forms grow. They contain phycoerythrin pigment which can absorb blue-green part of the sunlight spectrum which can penetrate deep into the water.

Economic Importance :

(i) **Photosynthesis :** At least a half of the total carbon dioxide fixation on earth is carried out by algae through photosynthesis.

(ii) **Primary producers :** Algae are primary producers of energy-rich compounds (food). This form the basis of food cycles of all aquatic animals.

(iii) **Food supplements :** Many species of *Porphyra* (Red algae), *Laminaria* and *Sargassum* (Brown algae) are among the 70 species of marine algae used as food. Certain unicellular algae like *Chlorella* and *Spirullina* are rich in proteins and are used as food supplements even by space travellers.

(iv) **Hydrocolloids :** Hydrocolloids are water-holding substances. **Algin** and **carrageen** are hydrocolloids obtained from brown and red algae, respectively. Agar (a hydrocolloid) is one of the commercial products obtained from *Gelidium* and *Gracilaria* and are used to grow microbes and in preparations of ice-creams and jellies.

(v) **Iodine :** *Fucus* and *Laminaria* are rich sources of iodine.

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BRYOPHYTES

Bryophytes are non-vascular terrestrial plants of moist habitats in which a multicellular diploid sporophyte lives as a parasite on an independent multicellular haploid gametophyte.

The main characteristics of bryophytes are :

1. They are found commonly in damp, humid and shaded localities.
2. The plants are small. They seldom attain great length or height.
3. They are also called **amphibians** of the plant kingdom because these plants can live in soil but are dependent on water for sexual reproduction.
4. The plant body is thallus-like and may be prostrate or erect. They lack true roots, stems or leaves but possess root-like, leaf-like or stem-like structures. The plant body is attached to the substratum by unicellular or multicellular root-like structures called rhizoids.
5. Vascular tissues *i.e.*, xylem and phloem are absent.
6. The dominant phase or the main plant body is a free-living gametophyte.
7. Gametophyte is the haploid stage of a plant that generates gametes by the process of mitosis, hence is called as gametophyte.
8. The gametophyte of bryophyte has multicellular sex organs. The male sex organ is called **antheridium** and the female sex organ is flask-shaped called **archegonium**.

   Antheridium is surrounded by a sterile jacket which encloses mass of sperm mother cells / androcytes which produce two biflagellated antherozoids (motile male gametes). Archegonium is flask-shaped with tubular neck and swollen venter. Just like antheridium, female sex organ is also surrounded by a jacket. Venter cavity possesses a sterile venter canal cell and a fertile egg. Neck encloses a few sterile neck canal cells.

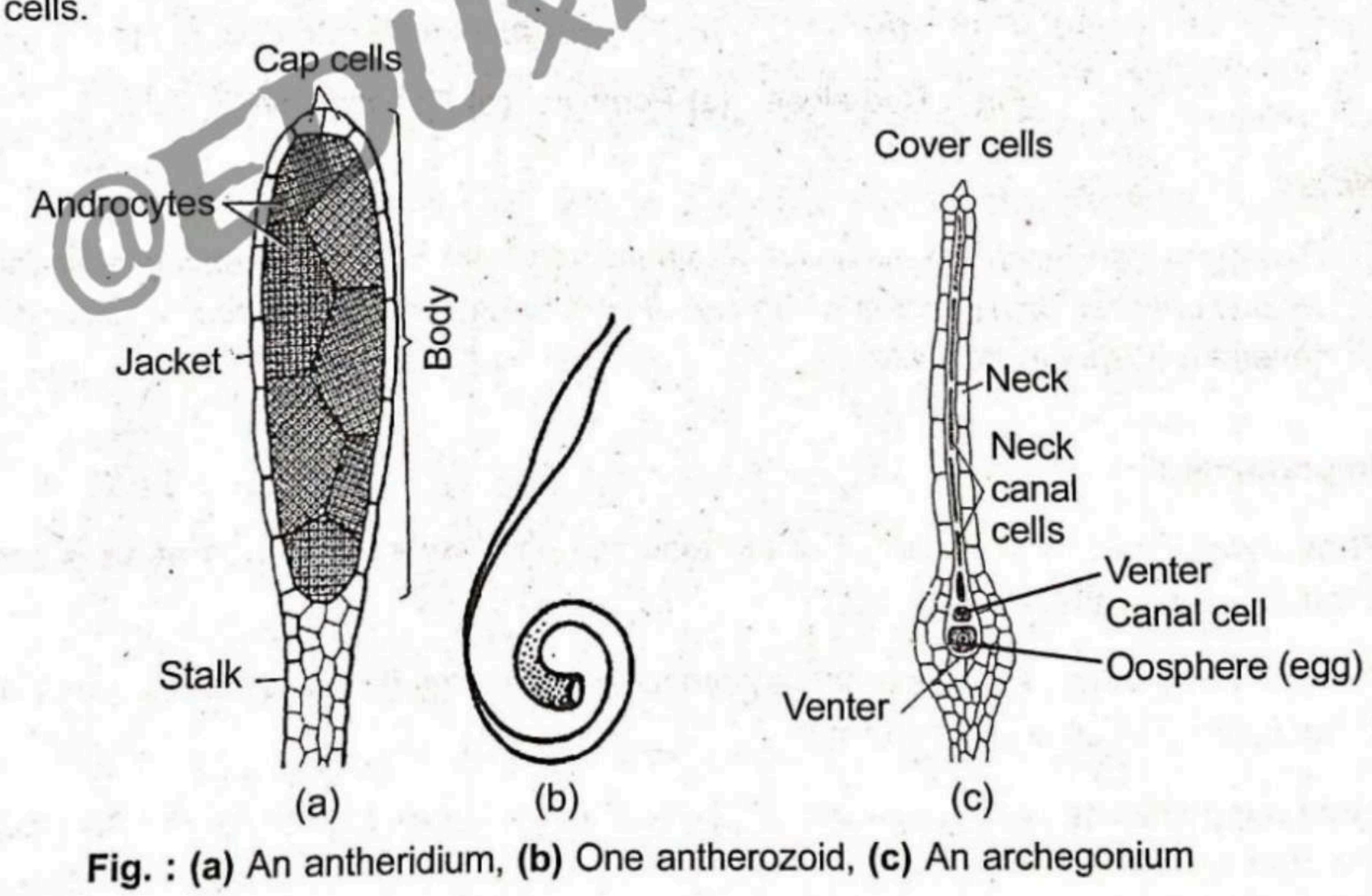


Fig. : (a) An antheridium, **(b)** One antherozoid, **(c)** An archegonium

9. An external layer of water is essential for the swimming of antherozoids to the archegonia.
10. Fertilisation produces zygote that is formed inside the archegonia. Zygotes do not undergo reduction division (meiosis) immediately instead they undergo mitotic division to form the embryo which develop further into diploid sporophyte. They are the first **embryophytes**.
11. The sporophyte is not free-living but attached to the photosynthetic gametophyte and derives nourishment from it.

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12. The sporophyte of bryophytes consists of three parts namely **capsule**, **seta** and **foot**. Inside the **capsule**, the **spore mother cells** undergoes meiosis to produce haploid spores (sporic meiosis).
13. Bryophytes are **homosporous** *i.e.*, they produce only one type of spores.
14. The spores get disseminated by wind. As the air shakes the capsule, the spores come out and are dispersed. Spores have the ability to germinate immediately after falling on the suitable substratum.
15 When the spores fall on a suitable substratum, they germinate either directly into the thalloid gametophyte (as in liverworts) or through a filamentous stage called protonema (as in mosses).
16. Vegetative reproduction occurs through fragmentation, gemmae and budding.

The bryophytes are divided into **liverworts** and **mosses**.

Liverworts

1. The plant body of liverwort is thalloid as in *Marchantia*. The thallus is dorsiventral and closely appressed to the substratum. The leafy members like *Porella* have tiny leaf-like appendages in two rows on the stem like structure.
2. Asexual reproduction occurs by means of fragmentation or by the formation of specialised structures called **gemmae** (sing. – gemma). Gemmae are green, multicellular asexual buds which develop in small receptacles called gemma cups located on the thalli. Mature gemmae become detached from the parent body and germinate to form new individual.
3. During sexual reproduction the sex organs are produced either on the same (*Riccia*) or on different thalli (*Marchantia*). In *Marchantia* sex organs are present on stalked receptacles. The male sex organs are present on **antheridiophore** and female sex organs are borne on the **archegoniophore**.

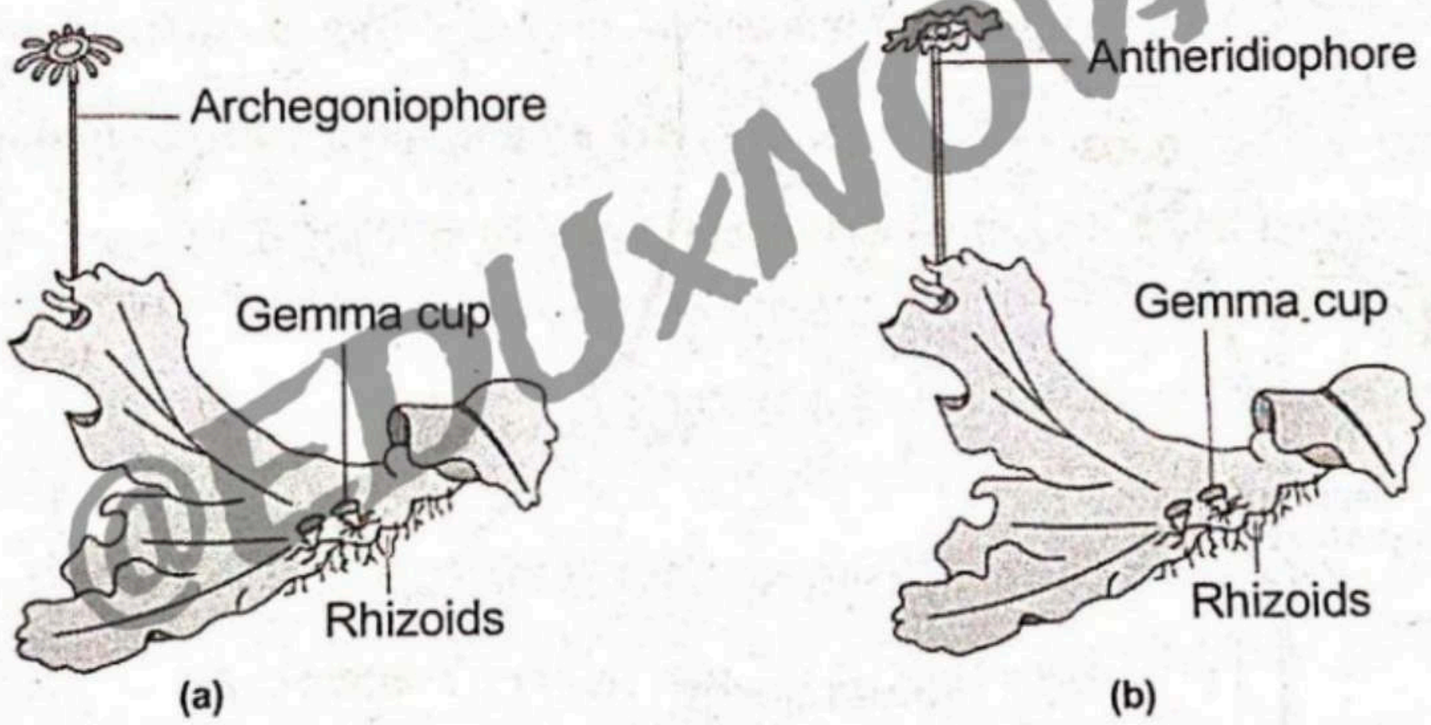


Fig. : Bryophytes : A liverwort – *Marchantia;* **(a)** Female thallus, **(b)** Male thallus

Mosses

1. The predominant stage of the life cycle of a moss is the gametophyte which consists of **protonema** and **leafy** stages.
2. The plant body is a leafy gametophyte which has multicellular and branched rhizoids. They consist of upright, slender axis bearing spirally arranged leaves. This stage bears the sex organs.
3. The sex organs antheridia and archegonia are produced at the apex of the leafy shoot. The sex organs are present on the same plant but on different branches.
4. The mosses have an elaborate mechanism of spore dispersal.
5. The spores on liberation germinate into a creeping, green, branched and frequently filamentous stage called **protonema** (juvenile stage).
6. Vegetative reproduction in mosses is by **fragmentation** and **budding** in the secondary protonema. Leafy stage develops from the secondary protonema as a lateral bud.

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7. Common examples of mosses are *Funaria*, *Polytrichum* and *Sphagnum*.

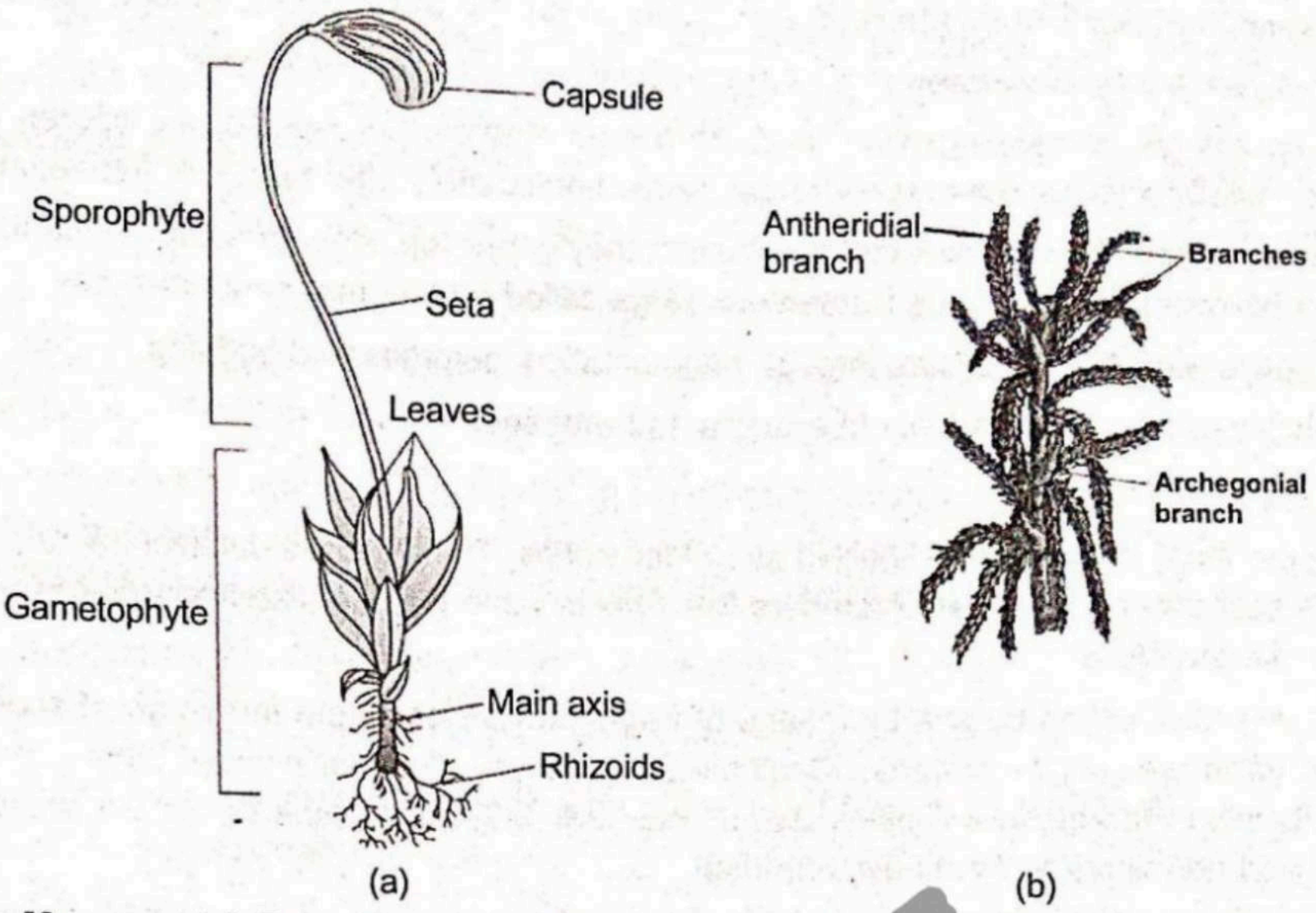


Fig. : Mosses : (a) *Funaria*, Gametophyte and Sporophyte, **(b)** *Sphagnum* Gametophyte

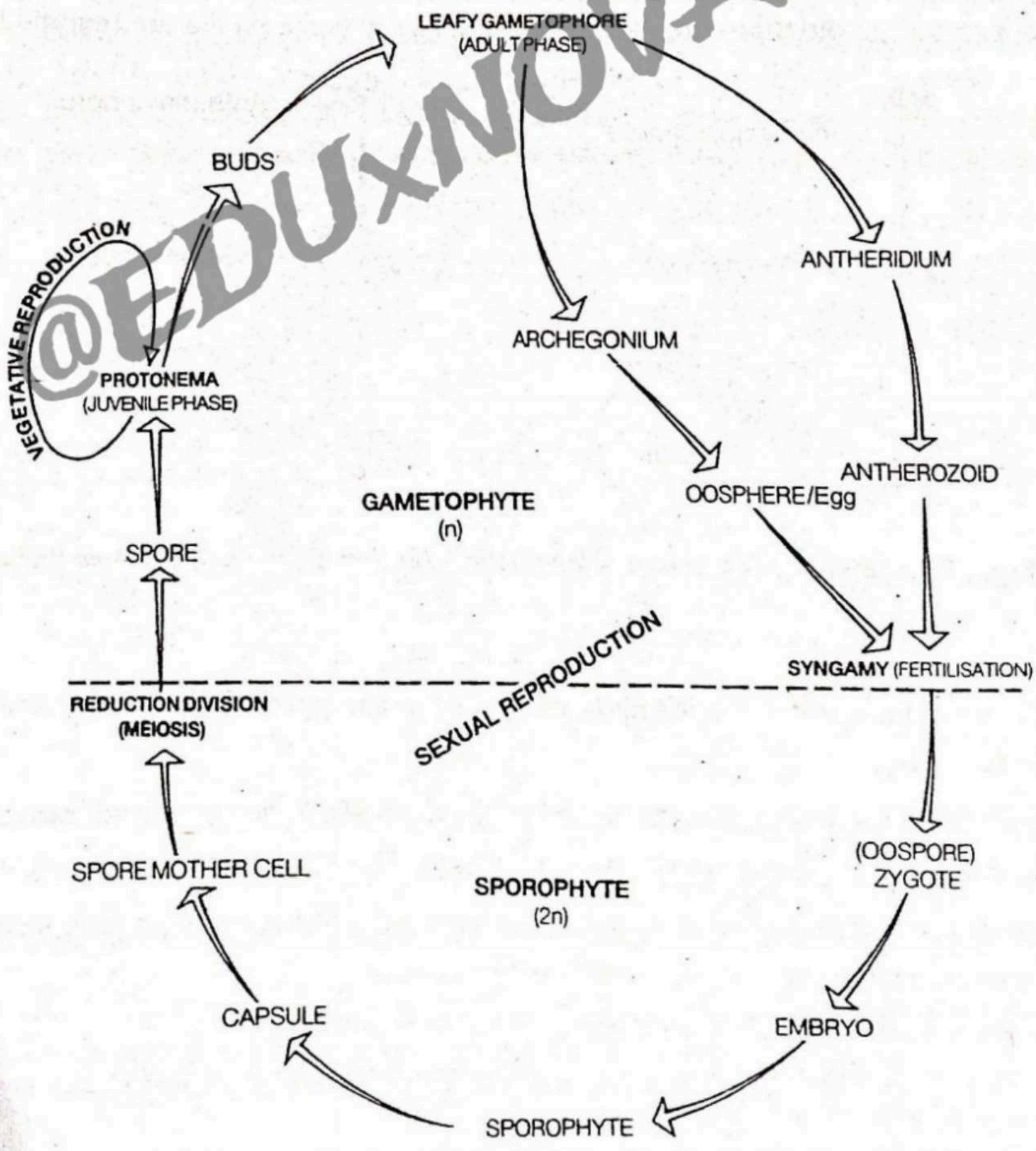


Fig. : Graphical Representation of Life Cycle of Moss

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Economic importance :

(i) **Prevention of soil erosion :** Bryophytes, especially mosses, form dense mats over the soil and prevent soil erosion against falling rains.

(ii) **Soil formation :** Mosses are an important link in plant succession on rocky areas. They take part in building soil in rock crevices formed by lichens. Growth of *Sphagnum* (Bog moss) ultimately fills ponds and lakes with soil.

(iii) **Water retention :** Dry *Sphagnum* has great water absorbing capacity. This characteristic is employed by gardeners to keep seedlings and cut plants moist during transportation and propagation. *Sphagnum* moss was used in place of absorbent cotton, so it is also called cotton moss.

(iv) **Peat :** *Sphagnum* often grows in acidic marshes. The older dead parts of moss and other marshy plants got slowly carbonised, compressed and fossilised over thousands of years and have produced a dark spongy mass called peat. Peat is dried, compressed and cut to form blocks. The peat blocks are used as fuel. Peat is also a good manure. It overcomes soil alkalinity and increases its water retention as well as aeration.

(v) **Other uses :** In rocky and ice cold areas mosses are a good source of food for certain animals like birds. A decoction of *Polytrichum commune* was employed in removing kidney stones.

PTERIDOPHYTES

They are seedless vascular plants including horsetails and ferns.

The main characteristics of pteridophytes are :

1. The pteridophytes are found in cool, damp, shady places though some may flourish well in sandy-soil conditions.
2. Evolutionarily, they are the first terrestrial plants to possess vascular tissues– xylem and phloem. Xylem transports water and minerals while phloem conducts organic food.
3. The dominant phase or independent plant is a sporophyte.
4. Main plant body is differentiated into true stem, leaves and roots.
5. The leaves in pteridophyta may be small (microphylls) as in *Selaginella* or large (macrophylls) as in ferns.

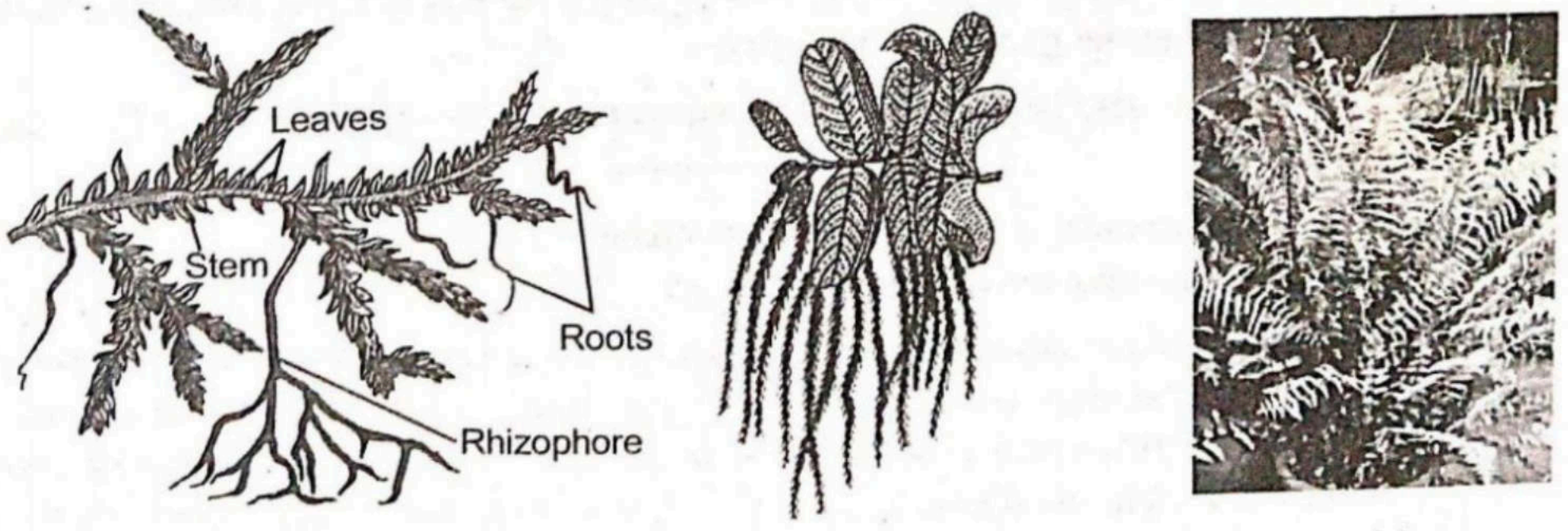


Fig. : (a) *Selaginella*, **(b)** *Salvinia* (aquatic fern), **(c)** Terrestrial fern

6. The leaflets and leaves having sporangia are called sporophylls.

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7. The sporangia contain sporogenous tissue, where spore mother cells undergo sporic meiosis to produce spores.
8. In some pteridophytes like *Selaginella* and *Equisetum* the sporophylls may be compactly arranged into structures called **strobili** or **cones**.

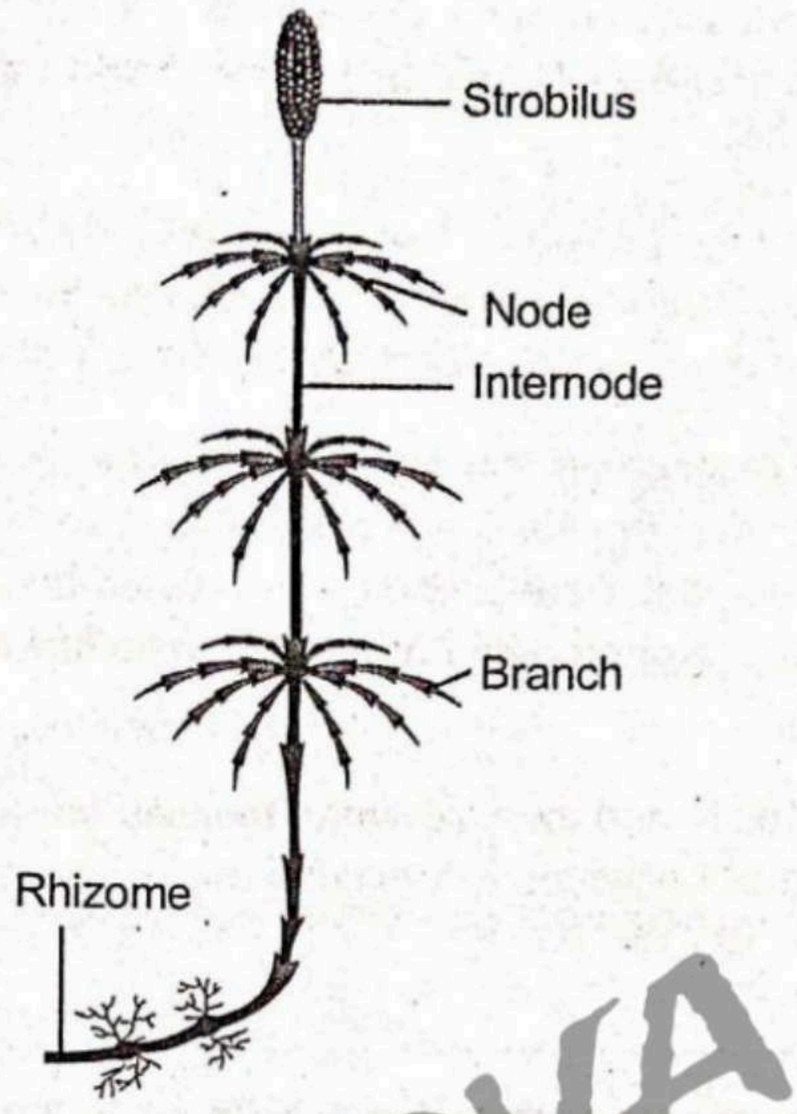


Fig. : *Equisetum*

9. Plants like *Lycopodium*, *Dryopteris* etc. produce a single kind of spores and are known as **homosporous** whereas plants like *Selaginella*, *Salvinia* etc. produce two kinds of spores, macro (large) and micro (small) spores. They are known as **heterosporous**.
10. The spores germinate to give rise to inconspicuous, small but multicellular free living, mostly photosynthetic thalloid gametophytes called **prothallus**.
11. The megaspores and microspores germinate and give rise to female and male gametophytes respectively.
12. The gametophytes bear male and female sex organs called antheridia and archegonia respectively.
13. The gametophyte that develops in homosporous species is **monoecious** while those of heterosporous species is **dioecious**.
14. These gametophytes require cool, damp, shady places to grow. Because of this specific restricted requirement and the need of water for fertilisation, **the spread of living pteridophytes is limited and restricted to narrow geographical regions.**
15. Sperms are flagellated (antherozoids). They require an external supply of water for swimming to the mouth of archegonium.
16. Fusion of gametes results in the formation of zygote which develop into embryo. The embryo develops into a multicellular well-differentiated sporophyte.
17. In the heterosporous species, the female gametophyte remains on the parent sporophytes for variable periods. The development of the zygotes into young embryos takes place within the female gametophytes. **This event is a precursor to the seed habit (tendency towards seed formation) considered an important step in evolution**. Hence, heterospory leads to seed habit in plants.
18. Heterosporous vascular cryptogams like *Selaginella* and *Marsilea* fail to develop seeds because:
    (a) They have no protective structure like the integuments surrounding the megasporangia.
    (b) The retention of megaspores permanently within the megasporangia has not become established.

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19. The pteridophytes are further classified into four classes : Psilopsida (*Psilotum*), Lycopsida (*Selaginella*, *Lycopodium*), Sphenopsida (*Equisetum*) and Pteropsida (*Dryopteris*, *Pteris*, *Adiantum*).

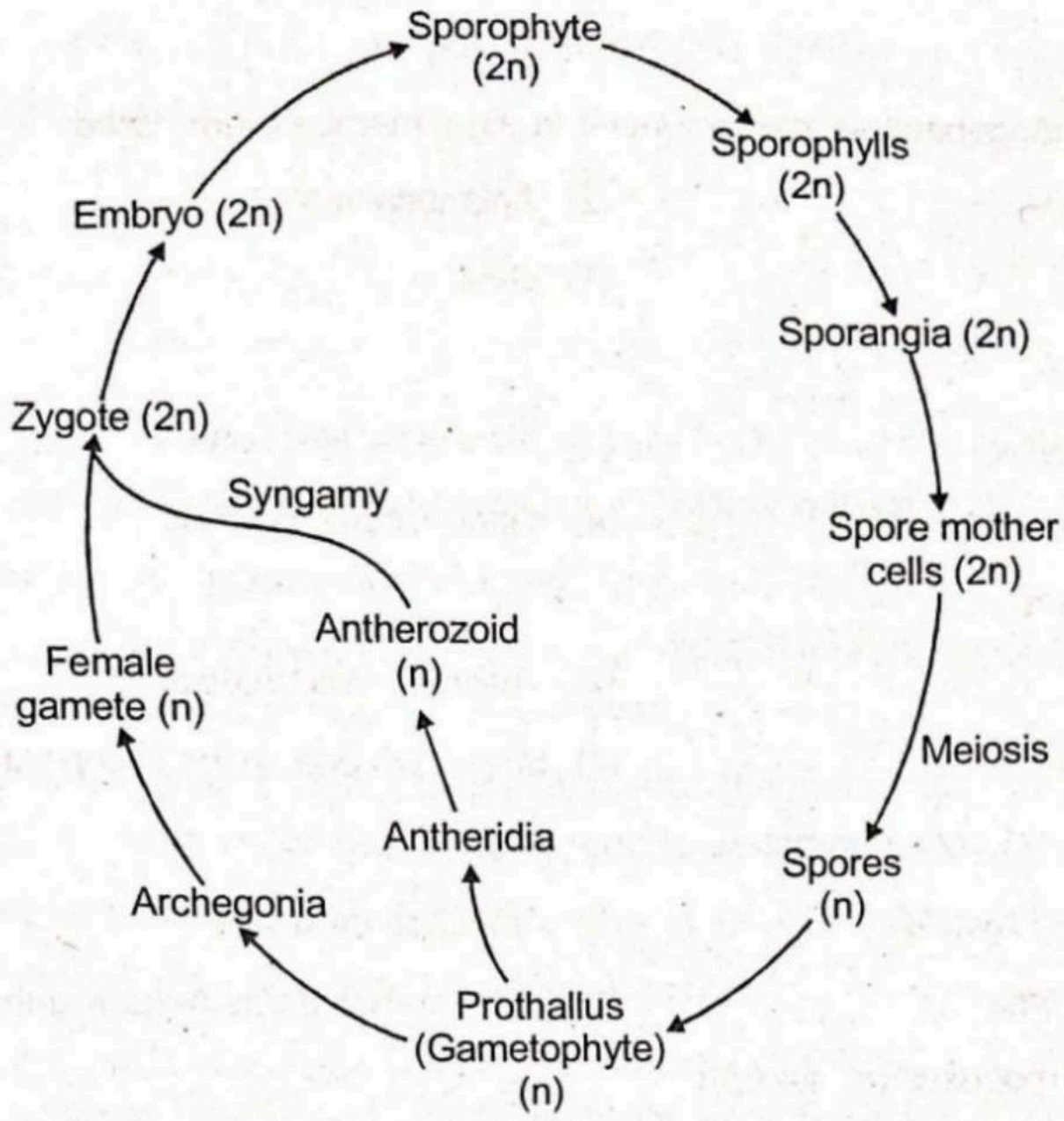


Fig. : Life cycle of a Fern plant

***Note* :**

Azolla is an aquatic water fern used as biofertilizer. The alga *Anabaena* fixes atmospheric nitrogen in symbiotic association with *Azolla*.

Economic importance :

(i) **Soil binding :** Pteridophytes bind the soil even along hill slopes. The soil is protected from erosion.

(ii) **Medicines :** An anthelmintic drug is obtained from a pteridophyte called *Dryopteris*.

(iii) **Ornamentals :** Ferns are grown as ornamental plants for their delicate and graceful leaves.

(iv) **Food :** *Marsilea*, a water fern yields starch that constitute a good source of food for certain tribals.

(v) **Scouring :** *Equisetum* stems have been used in scouring (cleaning of utensils) and polishing of metals.

EXERCISE

1. The artificial system of classification is based on

(1) One or few anatomical features (2) One or few morphological features

(3) Many morphological features (4) Many internal features

2. (A) Number and codes are assigned to few of the selected characters in numerical taxonomy.

(B) Phylogenetic classification systems are based on evolutionary relationships between the various organisms.

(C) Cytotaxonomy is based on cytological information like chromosome number, structure & behaviour.

(1) A and B are incorrect (2) Only C is correct

(3) Only A is incorrect (4) B and C are incorrect

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3. Which of the following is a colonial algae?

(1) *Chlamydomonas* (2) *Ulothrix*

(3) *Volvox* (4) *Spirogyra*

4. Motile, asexual and endogenous spores produced in algal members are called

(1) Zoospores (2) Aplanospores

(3) Conidia (4) Cyst

5. Bryophytes are

(1) Always homosporous (2) Always heterosporous

(3) Sometimes heterosporous (4) Seldom homosporous

6. Plant body of liverworts is

(1) Thalloid in *Porella* (2) Thalloid in *Marchantia*

(3) Leafy in *Marchantia* (4) More than one option is correct

7. Select incorrect statement w.r.t. characters of true moss.

(1) Multicelled branched rhizoids (2) Presence of scales

(3) Presence of protonema (4) Erect leafy axis as mature gametophyte

8. All given members are monoecious, *except*

(1) *Marchantia* (2) *Funaria*

(3) *Anthoceros* (4) *Sphagnum*

9. Pteridophytes are also called

(1) Vascular amphibians of plant kingdom (2) First tracheophytes

(3) Botanical snakes (4) All of these

10. A. Companion cells and sieve tubes are absent in pteridophytes.

B. Gametophyte of pteridophytes require cool, dry and shady places to grow.

C. Prothallus is found in *Dryopteris*

(1) Only C is correct (2) Only A is incorrect

(3) A and B are correct (4) Only B is incorrect

11. Spread of living pteridophytes is restricted to narrow geographical regions due to need for

(1) Water (2) Food

(3) Chemicals (4) More than one option is correct

12. True ferns are associated with

(1) Macrophylls (2) Microphylls

(3) Strobilus (4) Thalloid sporophyte

13. Select a set of heterosporous genera.

(1) *Marsilea, Azolla* (2) *Salvinia, Pteridium*

(3) *Adiantum, Azolla* (4) *Pteris, Lycopodium*

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14. *Dryopteris* is/has
(1) Gametophyte as main plant body
(2) Homosporous
(3) Non-motile male gametes
(4) Shows seed habit

15. _____ is used as biofertilizer :
(1) *Azolla*
(2) *Marsilea*
(3) *Equisetum*
(4) *Salvinia*

GYMNOSPERMS

The gymnosperms (gymnos : naked, sperma : seeds) are plants in which the seeds are not enclosed within fruit wall, *i.e.*, they bear naked seeds. These are seeded plants without fruits.

The main characteristics of gymnosperms are :

1. They include medium-sized trees or tall trees and shrubs. The *Sequoia* (red wood tree) is one of the tallest tree species smallest gymnosperm is ***Zamia***.
2. Unlike bryophytes and pteridophytes, in gymnosperms the male and female gametophytes do not have an independent free-living existence.
3. The plants represent the diploid sporophytic phase in the life cycle.
4. Plant body is differentiated into roots, stem and leaves.
5. The **roots** are generally **tap roots**. Roots in some genera (*Pinus*) have fungal association in the form of **Mycorrhiza**. While in some others (*Cycas*) small specialised roots called **coralloid roots** are associated with nitrogen-fixing cyanobacteria. Coralloid roots are irregular and do not possess root hairs and root caps.
6. The **stems** are unbranched (*Cycas*) or branched (*Pinus*, *Cedrus*).
7. The **leaves** may be simple or compound. In *Cycas* the **pinnate leaves** persist for a few years.

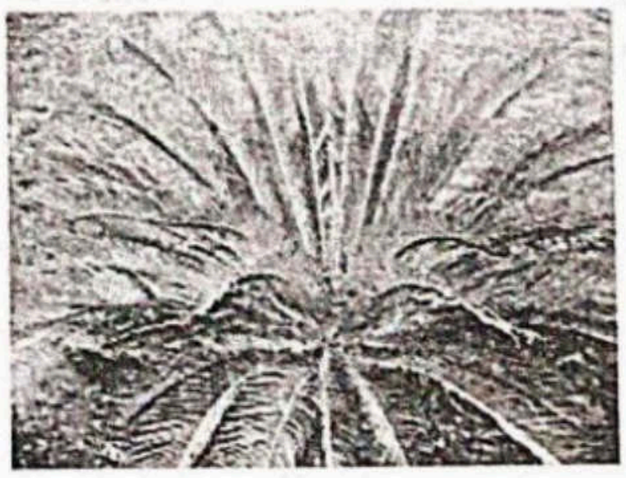


Fig. : *Cycas*

8. The leaves in gymnosperms are well-adapted to withstand extremes of temperature, humidity and wind. In conifers, the needle-like leaves reduce the surface area. Their thick cuticle and sunken stomata also help to reduce water loss. Leaves of *Ginkgo biloba* are fan-shaped.
9. They have well-developed vascular tissues – xylem and phloem.
10. Gymnosperms are heterosporous. The two kinds of spores are produced within sporangia that are borne on sporophylls which are arranged spirally along an axis to form lax or compact strobili or cones.
11. Sporophylls are of two types : microsporophylls and megasporophylls.
12. Microsporophylls are compactly arranged on a central axis to form a **male strobili** or microsporangiate/ pollen cone. They bear microsporangia where microspore mother cells undergo meiosis to form microspores. The microspores develop into a male gametophytic generation which is **highly reduced** (haploid) and is confined to only a limited number of cells. This reduced gametophyte is called a **pollen grain**.

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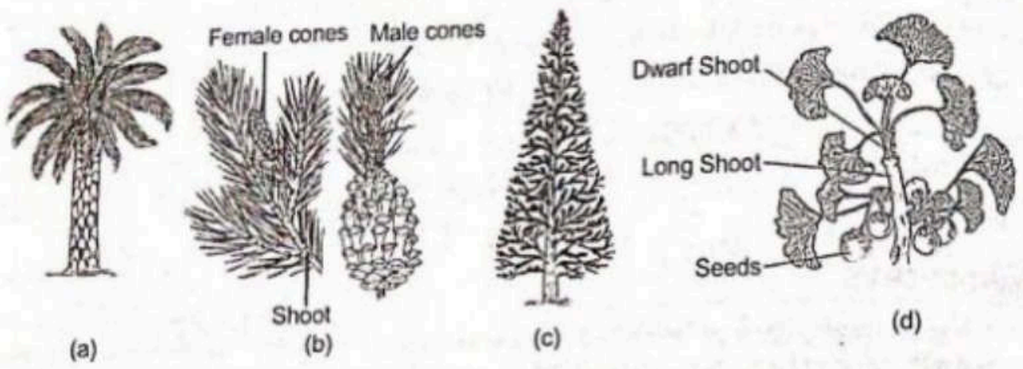


Fig. : Some Gymnosperms :
(a) *Cycas*, **(b)** *Pinus*, **(c)** *Cedrus*, **(d)** *Ginkgo*

13. Megasporophylls are compactly arranged to form **female strobili** or macrosporangiate/seed cone. They bear the integumented megasporangia (ovules). The ovule or the integumented megasporangium mainly consists of a body called nucellus protected by envelope or integument.

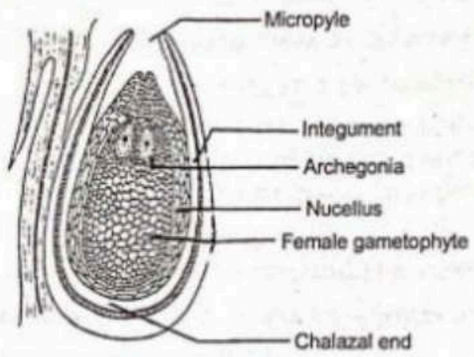


Fig. : Ovule or Integumented megasporangium

14. The male or female reproductive bodies may be borne on the same tree (*Pinus*) or on different trees (*Cycas*).
15. One of the cells of the nucellus differentiates as megaspore mother cell which undergoes meiosis to form a tetrad of megaspores of which three degenerate.
16. The functional megaspore enclosed within the megasporangium (nucellus) undergoes repeated mitosis and develops into a multicellular female gametophyte that bears two or more **archegonia** (female sex organs). The multicellular female gametophyte is also retained within megasporangium.
17. During pollination the pollen grains are carried in **air currents** and reach the opening of the ovules (micropylar end) borne on megasporophylls.
18. Each pollen grain germinates and the resulting pollen tube discharge the male gametes near the mouth of the archegonia.
19. Fertilisation results in the formation of zygote, which develops into an embryo.
20. Ovules become transformed into seeds. These seeds are not covered.
21. The three generations in seed are :
    (i) Testa, tegmen and perisperm represent parental sporophyte.
    (ii) Endosperm represents female gametophyte.
    (iii) Plumule, radicle, suspensor and cotyledons (embryo) represent future sporophyte.

22. **Polyembryony :** It is the formation of more than one embryo inside a single seed. It was reported by Leeuwenhoek in oranges. Simple polyembryony is due to fertilisation of many eggs, *e.g.*, *Pinus* ovule has 2-8 archegonia. Cleavage polyembryony is true polyembryony and very common. It is due to splitting of embryo tissue. Adventive polyembryony is the formation of extra embryos directly from diploid cells (*e.g.*, rosette cells) other than embryonal cells.

Note :

1. Order Gnetales consists of *Gnetum, Ephedra and Welwitschia*. These are nearer to flowering plants in having flower like arrangement of sporophyll and possessing primitive vessels in xylem so wood is called heteroxylous. Plants of Cycadales and Coniferales are commonly called Cycads and Conifers respectively.
2. *Cycas, Ginkgo* and *Metasequoia* are living fossils.
3. *Ginkgo biloba* (Pagoda tree *or Maiden hair tree) is oldest living fossil and it is connecting link between cycades and conifers.*

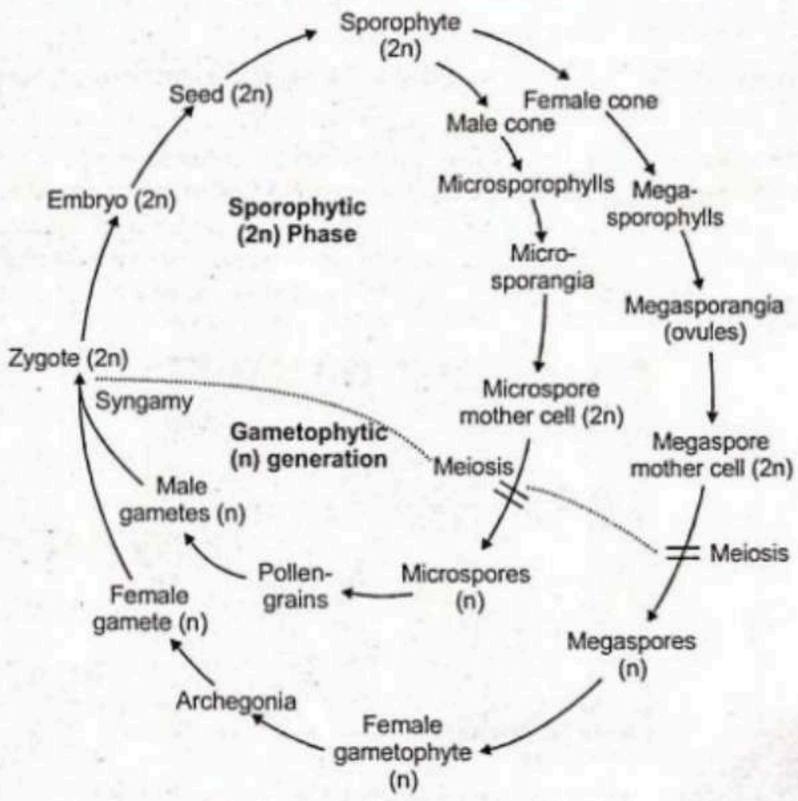


Fig: Life cycle of gymnosperm (*Pinus*)

Economic importance :

(i) Edible seeds : Seeds of *Pinus gerardiana* called chilgoza are edible. They are eaten after roasting.

(ii) Timber : Gymnosperms possess softwood. It is used in preparation of light furniture, plywood, packing cases, match-sticks, railway sleepers, etc.

(iii) Paper : A number of gymnosperm woods are used in the manufacture of paper.

(iv) Fibre boards : Needle of *Pinus* and other conifers are used in making fibre boards that are used in making packing cases.

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(v) **Ephedrine :** Drug ephedrine is obtained from *Ephedra*, a gymnosperm. The drug is used in curing respiratory ailments, including asthma.

(vi) **Taxol :** An anticancerous drug, called Taxol, is obtained from the bark of *Taxus*.

Note :

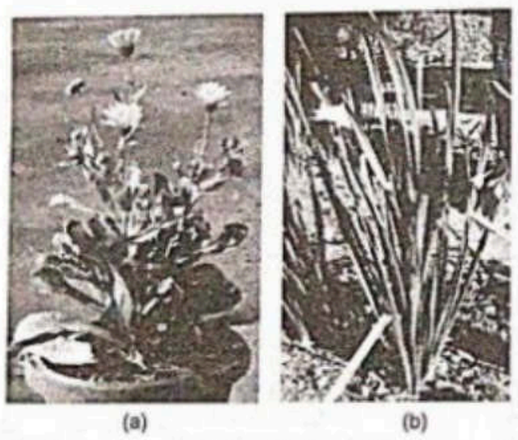
Sulphur shower : *Sulphur shower is a phenomenon related to reproduction/pollination in pine trees. Since Pinus bears cones, they are called conifers. Pines have leaves in the form of needles. The cones are of two types – seed cones (female) and pollen cones (male) on the same plant. As the pollen cones mature, they discharge large quantities of pollen grains into the air and it appear as a yellow cloud of yellow pollen grains. This is called sulphur shower and occurs during spring.*

ANGIOSPERMS

Angiosperms (Gk. *angion* – vessel, *sperma* – seed) are those plants in which seeds are formed inside fruit and the sporophylls are organised into **flowers.** Hence, they are called flowering plants.

The main characteristics of angiosperms are :

1. They range in size from tiny, almost microscopic *Wolfia* (smallest angiosperm) to tall trees of *Eucalyptus* (over 100 metres).
2. Angiosperms are divided into two classes – on the basis of number of embryonic leaves or cotyledons : the **dicotyledons** and the **monocotyledons**. The dicotyledons are characterised by seeds having two cotyledons, reticulate venations in leaves, and tetramerous or pentamerous flowers, i.e., having four or five members in each floral whorls. The monocotyledons on the other hand are characterised by single cotyledonous seeds, parallel venation in leaves, and trimerous flowers having three members in each floral whorls.



(a) (b)

Fig. : Angiosperms: **(a)** A dicotyledon, **(b)** A monocotyledon

3. Angiosperms are the most recently and highly evolved plants.
4. A flower is the shoot modified for reproduction. The male and female sex organs of a flower are stamen and pistil/carpel respectively.
5. A microsporophyll or stamen consists of a slender filament and an anther at the tip. Within the anthers, the pollen mother cell divides by meiosis to produce microspores which nature into pollen grains.
6. The female sex organ, Pistil consists of an ovary at its base, a lung slender style and stigma. Inside the ovary ovules are present. Generally each ovule has a megaspore mother cell that undergoes meiosis to produce haploid megaspores.

7. Three of them degenerate and one divide to form female gametophyte or embryo sac (7 celled and 8 nucleate) inside ovule.
8. Pollen grains or microspores reach stigma through pollination.
9. Archegonia are absent. Instead, there is one oosphere surrounded by two specialised synergid cells (egg apparatus) inside the embryo sac. Apart from egg apparatus, antipodals and central cell are also present.
10. There is **double fertilisation** (syngamy and triple fusion).
11. **Endosperm** is formed through **triple fusion** and is generally **triploid.** Embryo is formed by syngamy.
12. Fertilised ovules ripen into seeds . The seeds are covered by fruits. **A fruit is technically a ripened ovary.** Fruits not only protect the seeds but also help in their dispersal.

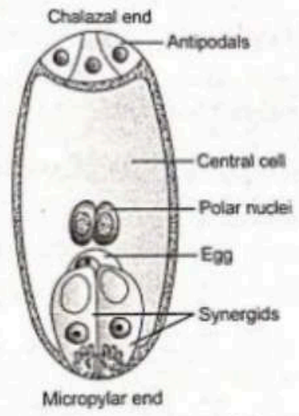


Fig. : A diagrammatic representation of the mature embryo sac.

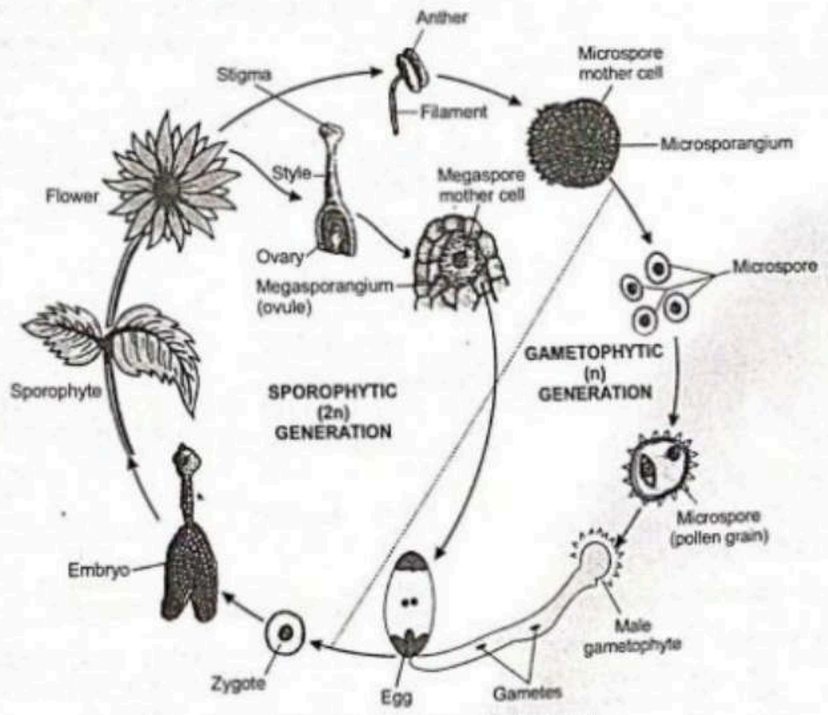


Fig. : Life cycle of an angiosperm

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Economic Importance :

(i) **Food :** Flowering plants are the major sources of food. They produce cereals such as wheat, rice, etc., vegetables such as potato, tomato, carrot, etc., and fruits such as apple, mango.

(ii) **Edible oils :** Flowering plants are the main source of edible oils used for cooking. These are obtained from groundnut, mustard, sunflower, coconut etc.

(iii) **Spices :** Common spices are obtained from various parts of flowering plants such as coriander, cloves, black pepper, etc.

(iv) **Beverages :** Coffee and cocoa are the common beverages obtained from the flowering plants.

(v) **Medicines :** Some of the important medicinal plants are quinine, belladonna, ashwagandha etc.

(vi) **Timber :** Many angiosperms, particularly dicotyledons, yield valuable hard wood which is used as timber *e.g.*, teak, oak, sisham, sandalwood etc.

(vii) **Fibres :** Fibres of different qualities are obtained from various species of flowering plants, *e.g.*, Textile fibres are obtained from cotton.

Scan the QR code for further explanation

PLANT LIFE CYCLES AND ALTERNATION OF GENERATIONS

During the life cycle of a sexually reproducing plant, there is an alternation of generations between gamete producing haploid gametophyte and spore producing diploid sporophyte life cycles patterns of different plant groups are:

1. Haplontic Life Cycle

(i) The dominant phase is a free-living photosynthetic haploid gametophyte.

(ii) The diploid or sporophytic generation is represented only by the single-celled zygote.

(iii) The zygote undergoes meiosis (zygotic meiosis) and forms haploid spores. The spores on germination give rise to gametophyte. e.g.,many algae such as *Volvox*, *Spirogyra* and some species of *Chlamydomonas*.

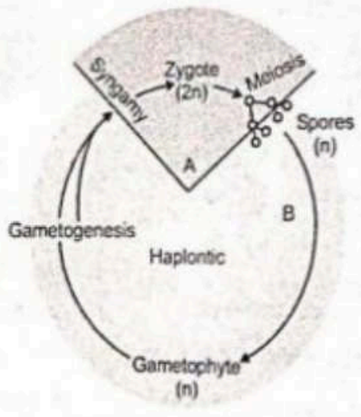


Fig. : Haplontic Life cycle

2. Diplontic Life Cycle

(i) The dominant phase is the photosynthetic independent diploid sporophytic plant.

(ii) Meiosis occurs in the spore mother cells (diploid) and the gametophytic phase is represented by the single to few celled haploid gametophyte which is short-lived and is dependent on the photosynthetic sporophyte e.g., all seed-bearing plants, *i.e.*, gymnosperms and angiosperms.

Fucus, a brown alga.

Fig. : Diplontic Life Cycle

3. Haplo-diplontic Life Cycle

(i) It is an intermediate type of life cycle, where both the sporophyte and gametophyte are multicellular, equally dominant and often free-living. Bryophytes and pteridophytes exhibit this pattern.

(ii) In bryophytes, the dominant phase is gametophyte and it alternates with the short-lived sporophyte.

(iii) In pteridophytes, the dominant phase is sporophyte. It alternates with short-lived haploid gametophyte.

Algae exhibit haplontic life cycle but some of them such as *Ectocarpus*, *Polysiphonia*, kelps are haplo-diplontic.

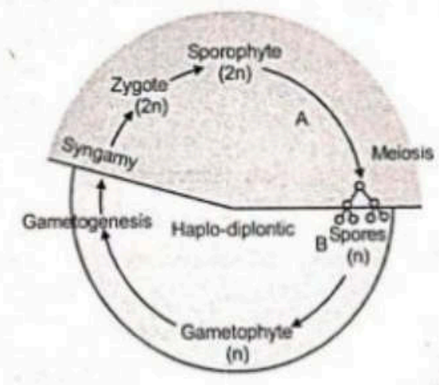


Fig. : Haplo-diplontic Life Cycle

EXERCISE

16. Endosperm in gymnosperms is

(1) n (2) 2n

(3) 3n (4) 4n

17. Pollen grain is

(1) Highly reduced male gametophyte

(2) Well developed male gametophyte

(3) Highly reduced sporophyte

(4) Well developed female gametophyte

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18. A Siphonogamy is found in *Pinus*.
B. Stem branches are monomorphic in *Pinus*.
(1) A and B are incorrect
(2) Only A is correct
(3) Only B is correct
(4) A and B are correct

19. The archegoniate spermatophytes are
(1) Angiosperms
(2) Bryophytes
(3) Pteridophytes
(4) Gymnosperms

20. Cyanobacteria is found in association with
(1) Coralloid roots of *Pinus*
(2) Mycorrhizal roots of *Pinus*
(3) Coralloid roots of *Cycas*
(4) Mycorrhizal roots of *Cycas*

21. Needle leaf of conifers is a
(1) Xerophytic adaptation
(2) Mesophytic adaptation
(3) Hydrophytic adaptation
(4) Halophytic adaptation

22. Select correct w.r.t. diplohaplontic life cycle.
(1) Found in *Polysiphonia* and *Gnetum*
(2) Both gametophyte and sporophyte phases are present
(3) Common in green algae
(4) Gametic meiosis occurs

23. The phanerogams with ovary are
(1) Angiosperms
(2) Gymnosperms
(3) Bryophytes
(4) Pteridophytes

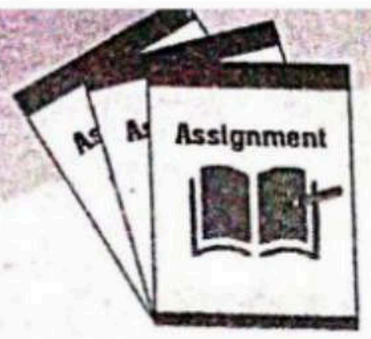
24. Fusion of male gamete with diploid secondary nucleus produces _____ and it is known as _____ (respectively).
(1) PEN; Triple fusion
(2) PEN; Syngamy
(3) Zygote; Syngamy
(4) Zygote; Triple fusion

25. A typical embryo sac is
(1) 8-nucleated and 7-celled
(2) 7-nucleated and 7-celled
(3) 8-nucleated and 8-celled
(4) 7-nucleated and 8-celled

❑ ❑ ❑

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Assignment



SECTION - A

NCERT Based MCQs

1. Life cycle of *Polysiphonia* and *Chlamydomonas* are respectively **[NCERT Pg. 42, 43]**

   (1) Haplontic and diplontic

   (2) Diplontic and haplodiplontic

   (3) Haplodiplontic and diplontic

   (4) Haplodiplontic and haplontic

2. All seed bearing plants have/show **[NCERT Pg. 38-40]**

   (1) Meiosis in zygote

   (2) Single to few celled haploid gametophytes

   (3) Multicellular and independent gametophytes

   (4) Alternation of dominant gametophyte with short lived multicellular sporophyte

3. State true **(T)** or false **(F)** for the given statements and choose the **correct** option **[NCERT Pg. 40, 41]**

   a. *Eucalyptus* is one of the tallest plant species

   b. Primary endosperm nucleus (PEN) of angiosperms is a triploid structure

   c. Embryo sac is a 3-celled structure

	a	b	c
(1)	T	T	T
(2)	F	F	T
(3)	T	T	F
(4)	F	T	F

4. Coralloid roots **[NCERT Pg. 38]**

   (1) Have symbiotic association with *Rhizobium* for N_2-fixation

   (2) Have fungal association

   (3) Are associated with N_2-fixing cyanobacteria

   (4) Are found in *Pinus*

5. Double fertilisation is an event unique to **[NCERT Pg. 41]**

   (1) Gymnosperms (2) Angiosperms

   (3) Pteridophytes (4) Bryophytes

6. Read the following statements **[NCERT Pg. 38, 39]**

   a. Ovules are not enclosed by ovary wall.

   b. Leaves are well adapted to withstand extremes of temperature, humidity & wind.

   c. Embryo sac represents the female gametophyte

   d. The ovules are borne on megasporophylls which may be clustered to form the female cone.

   How many of the above statements are **correct** regarding gymnosperms?

   (1) One (2) Two

   (3) Three (4) Four

7. Which one of the following pair is **wrongly** matched? **[NCERT Pg. 39]**

   (1) Gymnosperms – Homosporous plants

   (2) *Cycas* – Pinnate leaves

   (3) Angiosperms – Non-archegoniate phanerogams

   (4) *Sequoia* – Tallest gymnosperm

8. Which of the following plants belongs to the class Pteropsida? **[NCERT Pg. 38]**

   (1) *Equisetum* (2) *Adiantum*

   (3) *Selaginella* (4) *Psilotum*

9. In pteridophytes, the main plant body is **[NCERT Pg. 36]**

   (1) Thalloid gametophyte

   (2) Thalloid sporophyte

   (3) Gametophyte which is differentiated into root, stem & leaves

   (4) Sporophyte which is differentiated into root, stem & leaves

10. Bryophytes are called amphibians of plant kingdom because **[NCERT Pg. 35]**

    (1) They play an important role during succession on bare rocks

    (2) They lack roots, stems or leaves

    (3) They are dependent on water for sexual reproduction

    (4) They are more differentiated than that of algae

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11. Retention of female gametophyte in megasporangia is the event precursor to the seed habit which first appeared in **[NCERT Pg. 38]**
(1) Bryophytes (2) Pteridophytes
(3) Gymnosperms (4) Angiosperms

12. Rhizoids in mosses are **[NCERT Pg. 36]**
(1) Multicellular & branched
(2) Multicellular & unbranched
(3) Unicellular & elongated
(4) Unicellular & branched

13. Which of the following is a **correct** statement? **[NCERT Pg. 35]**
(1) Spores germinate to produce dorsiventral thallus in mosses
(2) Spores germinate to give rise to protonema in ferns
(3) Gemmae are non-green, unicellular structures of liverworts
(4) Mosses along with lichens are the first organisms to colonise rocks

14. Which of the following provides 'peat'? **[NCERT Pg. 35]**
(1) *Sargassum* (2) *Sphagnum*
(3) *Spirulina* (4) *Polytrichum*

15. Artificial system of classification is based on **[NCERT Pg. 29]**
(1) A few observable characteristics
(2) Natural affinities among organisms
(3) Evolutionary relationships
(4) Fossil evidences

16. Which of the following is a colonial & motile alga? **[NCERT Pg. 32]**
(1) *Chara*
(2) *Porphyra*
(3) *Volvox*
(4) *Chlamydomonas*

17. Oogamous type of sexual reproduction and diplontic life cycle is seen in **[NCERT Pg. 42]**
(a) *Ectocarpus* (b) *Spirogyra*
(c) *Wolfia* (d) *Fucus*
(1) (a) and (d) (2) (b) and (c)
(3) (a) and (b) (4) (c) and (d)

18. Which of the following is **not** a filamentous alga? **[NCERT Pg. 32]**
(1) *Spirogyra*
(2) *Dictyota*
(3) *Chlamydomonas*
(4) *Laminaria*

19. In which of the following groups, motile gametes are absent? **[NCERT Pg. 30]**
(1) Green algae
(2) Mosses
(3) Brown algae
(4) Red algae

20. All of the following have macrophyllous leaves, **except** **[NCERT Pg. 36]**
(1) *Lycopodium* (2) *Adiantum*
(3) *Pteris* (4) *Dryopteris*

21. Photosynthetic pigments predominantly found in brown algae are **[NCERT Pg. 32]**
(1) Chl-a, b (2) Chl-b, c
(3) Chl-a, c (4) Chl-a, d

22. Prothallus of *Dryopteris* is **[NCERT Pg. 38]**
a. Inconspicuous b. Unicellular
c. Diploid d. Non-photosynthetic
e. Free-living f. Monoecious
(1) Only a, b and d (2) Only c, d and e
(3) Only b, c and d (4) Only a, e and f

23. In gymnosperms, pollen grains are carried to the opening of ovules by **[NCERT Pg. 39]**
(1) Insects (2) Air current
(3) Water current (4) Higher animals

24. How many of the following plant groups show(s) formation of embryo? **[NCERT Pg. 35,38,39,41]**
Algae, Bryophytes, Pteridophytes, Angiosperms
(1) 3 (2) 2
(3) 1 (4) 4

25. Floridean starch is stored food material in **[NCERT Pg. 33]**
(1) *Chara*
(2) *Porphyra*
(3) *Ectocarpus*
(4) *Cedrus*

SECTION - B

Objective Type Questions

1. Natural classification system is better than artificial system due to all the given reasons, **except**
(1) Large number of characters used
(2) Based on natural affinities
(3) Based on external features only
(4) Plant identification is easy

2. Choose the **correct** option w.r.t. *Funaria*.

	Dominant phase	Alternate phase
(1)	Short lived	Long lived
(2)	Form spores	Form gametes
(3)	Diploid	Haploid
(4)	Gametophyte	Sporophyte

3. Red algae reach the maximum depth in sea water where no other type of photosynthetic organisms grow. This is due to the abundant formation of
(1) Chlorophyll a (2) Phycocyanin
(3) Phycoerythrin (4) Chlorophyll d

4. Artificial classification system based on androecium structure was given by
(1) Linnaeus
(2) Bentham and Hooker
(3) Takhtajan
(4) Eichler

5. True heterospory was first seen in plant group which is
(1) Non-vascular embryophytes
(2) Vascular Cryptogams
(3) Phanerogams without ovary
(4) Spermatophytes

6. Non-archegoniates embryophytes are
(1) Algae
(2) Bryophytes
(3) Angiosperms
(4) Gymnosperms

7. Life cycle of *Ficus* is
(1) Haplontic
(2) Diplontic
(3) Haplo-diplontic
(4) Diplo-haplontic

8. Sporophytic generation is represented by single cell in
(1) *Chlamydomonas*
(2) Kelps
(3) *Ectocarpus*
(4) *Polysiphonia*

9. Bryophytes show resemblance with algae in
(a) Formation of motile sperms
(b) Absence of vascular tissues
(c) Presence of haploid gametophyte
(1) (a) & (b) only
(2) (a) & (c) only
(3) (b) & (c) only
(4) All (a), (b) and (c)

10. Which of the following is correct sequence of steps involved in life cycle of liverworts if we start it from its dominant phase?
[A - Meiosis, B - Gametogenesis, C - Gametophyte, D - Syngamy, E - Zygote, F - Sporophyte, G - Spore]
(1) F → A → G → D → B → C → E
(2) F → A → G → C → E → B → D
(3) C → D → E → B → F → A → G
(4) C → B → D → E → F → A → G

11. Select the incorrect match.
(1) Covered seed - *Cedrus*
(2) Tallest gymnosperm - *Sequoia*
(3) Branched stem - *Pinus*
(4) Pinnate leaves - *Cycas*

12. Consider the following statements w.r.t mosses and select the **correct** option.
a. The sporophyte is less elaborate than that of liverworts.
b. Zygote develops into green filamentous structure.
c. Sex organs are produced at the apex of the leafy shoots.
d. Leafy stage is developed from secondary protonema.
(1) a, b & c are correct
(2) Only a is incorrect
(3) c and d are correct
(4) Only d is correct

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13. In liverworts, gemmae are
(1) Non-green, multicellular vegetative structures
(2) Green, unicellular, asexual buds
(3) Non-green, unicellular vegetative buds
(4) Green, multicellular asexual buds

14. Which of the following brown algae have massive plant bodies?
(1) *Laminaria* and *Ectocarpus*
(2) *Sargassum* and *Laminaria*
(3) *Macrocystis* and *Laminaria*
(4) *Dictyota* and *Fucus*

15. In which of the following the female gametophyte is retained on the parent sporophytes for variable periods?
(1) First embryophytes
(2) Non tracheophytes
(3) Homosporous pteridophytes
(4) Heterosporous pteridophytes

16. Which of the following feature(s) of *Pinus* help to tolerate or withstand extreme conditions?
(1) Sunken stomata
(2) Pinnate leaves
(3) Needle-like leaves
(4) Both (1) and (3)

17. Select the common feature amongst *Ginkgo*, *Selaginella* and *Sphagnum*.
(1) Independent sporophyte
(2) Vascular tissues
(3) Flask-shaped female sex organ
(4) Independent gametophyte

18. Which of the following are heterosporous with dependent male and female gametophyte?
(1) *Adiantum*, *Selaginella* and *Salvinia*
(2) *Cycas*, *Pinus* and *Ginkgo*
(3) *Cedrus*, *Selaginella* and *Adiantum*
(4) *Salvinia*, *Cycas* and *Dryopteris*

19. Chemotaxonomy is based on
(1) DNA sequence
(2) Chemical nature of protein
(3) Chromosome structure
(4) Both (1) & (2)

20. Which of the following event is unique to angiosperms?
(1) Seeds dispersal
(2) Triple fusion
(3) Pollination
(4) Heterospory

21. Complexity in body organisation is maximum in 'a' and complex post-fertilization changes are seen in 'b'.
(1) a – Brown algae; b – Red algae
(2) a – Red algae; b – Red algae
(3) a – Green algae; b – Red algae
(4) a – Brown algae; b – Green algae

22. Similarity among characters of different organisms is determined by using various statistical formula in
(1) Chemotaxonomy (2) Numerical taxonomy
(3) Cytotaxonomy (4) Karyotaxonomy

23. Agar producing genera of rhodophyceae are
(1) *Fucus*, *Laminaria*
(2) *Chlorella*, *Spirulina*
(3) *Gelidium*, *Gracilaria*
(4) *Porphyra*, *Ectocarpus*

24. Which of the following group of plants are collectively called archegoniates?
(1) A + B
(2) B + P + G
(3) Only B + P
(4) A + B + P

A – Algae P – Pteridophytes
B – Bryophytes G – Gymnosperms

25. One of the reasons of failure of development of seeds in *Selaginella* is
(1) Formation of indehiscent megasporangium
(2) Absence of heterospory
(3) Formation of unprotected megasporangium
(4) No shedding of megaspore

26. Homosporous plant is
(1) *Marsilea*
(2) *Azolla*
(3) *Lycopodium*
(4) *Salvinia*

27. Gymnosperms are more advanced than pteridophytes. Which of the following features of gymnosperms, justify this statement?
(1) Naked seeds
(2) Non-motile female gamete
(3) Presence of vascular tissues
(4) Main body is sporophytic

28. Pollen tube is carrier of non-motile male gamete in
(1) *Cycas* (2) *Sphagnum*
(3) *Ficus* (4) *Fucus*

29. Dioecious liverwort is
(1) *Marchantia* (2) *Pteris*
(3) *Funaria* (4) *Selaginella*

30. Which of the following is **not** a feature of pteridophytes?
(1) Water is required for transfer of antherozoids
(2) Independent sporophyte
(3) Undifferentiated vascular tissues
(4) True roots

SECTION - C

Previous Years Questions

1. Match List-I with List-II **[NEET-2022 (Phase-2)]**

List-I	List-II
(a) *Chlamydomonas*	(i) Moss
(b) *Cycas*	(ii) Pteridophyte
(c) *Selaginella*	(iii) Alga
(d) *Sphagnum*	(iv) Gymnosperm

Choose the correct answer from the options given below
(1) (a) - (ii), (b) - (iii), (c) - (i), (d) - (iv)
(2) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)
(3) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
(4) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

2. Which of the following is incorrectly matched? **[NEET-2022 (Phase-1)]**
(1) *Porphyra* - Floridian Starch
(2) *Volvox* - Starch
(3) *Ectocarpus* - Fucoxanthin
(4) *Ulothrix* - Mannitol

3. Hydrocolloid carrageen is obtained from: **[NEET-2022 (Phase-1)]**
(1) Rhodophyceae only
(2) Phaeophyceae only
(3) Chlorophyceae and Phaeophyceae
(4) Phaeophyceae and Rhodophyceae

4. Match the plant with the kind of life cycle it exhibits: **[NEET-2022 (Phase-1)]**

List-I	List-II
(a) *Spirogyra*	(i) Dominant diploid sporophyte vascular plant, with highly reduced male or female gametophyte
(b) Fern	(ii) Dominant haploid free-living gametophyte
(c) *Funaria*	(iii) Dominant diploid sporophyte alternating with reduced gametophyte called prothallus
(d) *Cycas*	(iv) Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte

Choose the correct answer from the options given below :
(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(2) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

5. Gemmae are present in **[NEET-2021]**
(1) Some Liverworts (2) Mosses
(3) Pteridophytes (4) Some Gymnosperms

6. Which of the following algae produce Carrageen? **[NEET-2021]**
(1) Blue-green algae (2) Green algae
(3) Brown algae (4) Red algae

7. Which of the following algae contains mannitol as reserve food material? **[NEET-2021]**
(1) *Ulothrix*
(2) *Ectocarpus*
(3) *Gracilaria*
(4) *Volvox*

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8. Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as: **[NEET-2021]**

(1) Heterosporous (2) Homosorus
(3) Heterosorus (4) Homosporous

9. Floridean starch has structure similar to **[NEET-2020 (Phase-1)]**

(1) Amylopectin and glycogen
(2) Mannitol and algin
(3) Laminarin and cellulose
(4) Starch and cellulose

10. Which of the following pairs is of unicellular algae? **[NEET-2020 (Phase-1)]**

(1) *Gelidium and Gracilaria*
(2) *Anabaena and Volvox*
(3) *Chlorella and Spirulina*
(4) *Laminaria and Sargassum*

11. Strobili or cones are found in **[NEET-2020 (Phase-1)]**

(1) *Pteris* (2) *Marchantia*
(3) *Equisetum* (4) *Salvinia*

12. The plant parts which consist of two generations - one within the other **[NEET-2020 (Phase-1)]**

(a) Pollen grains inside the anther
(b) Germinated pollen grain with two male gametes
(c) Seed inside the fruit
(d) Embryo sac inside the ovule

(1) (a), (b) and (c)
(2) (c) and (d)
(3) (a) and (d)
(4) (a) only

13. Male and female gametophytes do not have an independent free living existence in: **[NEET-2020 (Phase-2)]**

(1) Bryophytes
(2) Pteridophytes
(3) Algae
(4) Angiosperms

14. Phycoerythrin is the major pigment in **[NEET-2020 (Phase-2)]**

(1) Brown algae (2) Red algae
(3) Blue green algae (4) Green algae

15. Which of the following statements is incorrect about gymnosperms? **[NEET-2020 (Phase-2)]**

(1) Their seeds are not covered
(2) They are heterosporous
(3) Male and female gametophytes are free living
(4) Most of them have narrow leaves with thick cuticle

16. From evolutionary point of view, retention of the female gametophyte with developing young embryo on the parent sporophyte for some time, is first observed in **[NEET-2019]**

(1) Liverworts (2) Mosses
(3) Pteridophytes (4) Gymnosperms

17. Which one is wrongly matched? **[NEET-2018]**

(1) Uniflagellate gametes – *Polysiphonia*
(2) Biflagellate zoospores – Brown algae
(3) Unicellular organism – *Chlorella*
(4) Gemma cups – *Marchantia*

18. Winged pollen grains are present in **[NEET-2018]**

(1) Mustard (2) *Cycas*
(3) *Pinus* (4) Mango

19. Which of the following statements is correct? **[NEET-2018]**

(1) Ovules are not enclosed by ovary wall in gymnosperms
(2) *Selaginella* is heterosporous, while *Salvinia* is homosporous
(3) Stems are usually unbranched in both *Cycas* and *Cedrus*
(4) Horsetails are gymnosperms

20. Select the mismatch : **[NEET-2017]**

(1) *Pinus* – Dioecious
(2) *Cycas* – Dioecious
(3) Salvinia – Heterosporous
(4) Equisetum – Homosporous

21. An example of colonial alga is **[NEET-2017]**

(1) *Chlorella*
(2) *Volvox*
(3) *Ulothrix*
(4) *Spirogyra*

22. Life cycle of *Ectocarpus* and *Fucus* respectively are **[NEET-2017]**

(1) Haplontic, Diplontic
(2) Diplontic, Haplodiplontic
(3) Haplodiplontic, Diplontic
(4) Haplodiplontic, Haplontic

23. Conifers are adapted to tolerate extreme environmental conditions because of **[NEET-2016 (Phase-2)]**

(1) Broad hardy leaves (2) Superficial stomata
(3) Thick cuticle (4) Presence of vessels

24. Which one of the following statements is **wrong**? **[NEET-2016 (Phase-2)]**

(1) Algae increase the level of dissolved oxygen in the immediate environment
(2) Algin is obtained from red algae, and carrageenan from brown algae
(3) Agar-agar is obtained from *Gelidium* and *Gracilaria*
(4) *Laminaria* and *Sargassum* are used as food

25. Select the **correct** statement **[NEET-2016]**

(1) The leaves of gymnosperms are not well adapted to extremes of climate
(2) Gymnosperms are both homosporous and heterosporous
(3) *Salvinia*, *Ginkgo* and *Pinus* all are gymnosperms
(4) *Sequoia* is one of the tallest trees

26. Which one is a **wrong** statement? **[Re-AIPMT-2015]**

(1) Brown algae have chlorophyll a and c, and fucoxanthin
(2) Archegonia are found in Bryophyta, Pteridophyta and Gymnosperms
(3) *Mucor* has biflagellate zoospores
(4) Haploid endosperm is typical feature of gymnosperms

27. Read the following five statements (A to E) and select the option with **all correct** statements **[AIPMT-2015]**

(A) Mosses and Lichens are the first organisms to colonise a bare rock.
(B) *Selaginella* is a homosporous pteridophyte.
(C) Coralloid roots in *Cycas* have VAM.
(D) Main plant body in bryophytes is gametophytic, whereas in pteridophytes it is sporophytic.
(E) In gymnosperms, male and female gametophytes are present within sporangia located on sporophyte.

(1) (B), (C) and (E) (2) (A), (C) and (D)
(3) (B), (C) and (D) (4) (A), (D) and (E)

28. In which of the following gametophyte is **not** independent free living? **[AIPMT-2015]**

(1) *Pinus* (2) *Funaria*
(3) *Marchantia* (4) *Pteris*

29. Which one of the following statements is **wrong**? **[AIPMT-2015]**

(1) Mannitol is stored food in Rhodophyceae
(2) Algin and carrageen are products of algae
(3) Agar-agar is obtained from *Gelidium* and *Gracilaria*
(4) *Chlorella* and *Spirulina* are used as space food

30. Male gametes are flagellated in: **[AIPMT-2015]**

(1) *Spirogyra* (2) *Polysiphonia*
(3) *Anabaena* (4) *Ectocarpus*

31. Which of the following is responsible for peat formation? **[AIPMT-2014]**

(1) *Marchantia* (2) *Riccia*
(3) *Funaria* (4) *Sphagnum*

32. Male gametophyte with least number of cells is present in **[AIPMT-2014]**

(1) *Pteris* (2) *Funaria*
(3) *Lilium* (4) *Pinus*

33. Which one of the following shows isogamy with non-flagellated gametes? **[AIPMT-2014]**

(1) *Sargassum* (2) *Ectocarpus*
(3) *Ulothrix* (4) *Spirogyra*

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34. Select the **wrong** statement **[NEET-2013]**

(1) Anisogametes differ either in structure, function or behaviour

(2) In Oomycetes female gamete is smaller and motile, while male gamete is larger and non-motile

(3) *Chlamydomonas* exhibits both isogamy and anisogamy and *Fucus* shows oogamy

(4) Isogametes are similar in structure, function and behaviour

35. Monoecious plant of *Chara* shows occurrence of **[NEET-2013]**

(1) Stamen and carpel on the same plant

(2) Upper antheridium and lower oogonium on the same plant

(3) Upper oogonium and lower antheridium on the same plant

(4) Antheridiophore and archegoniophore on the same plant

36. Read the following statements (A-E) and answer the question which follows them

(A) In liverworts, mosses and ferns gametophytes are free-living.

(B) Gymnosperms and some ferns are heterosporous.

(C) Sexual reproduction in *Fucus*, *Volvox* and *Albugo* is oogamous.

(D) The sporophyte in liverworts is more elaborate than that in mosses.

(E) Both, *Pinus* and *Marchantia* are dioecious.

How many of the above statements are **correct**? **[NEET-2013]**

(1) Two (2) Three

(3) Four (4) One

37. Which one of the following pairs is wrongly matched? **[AIPMT (Mains)-2012]**

(1) *Ginkgo* – Archegonia

(2) *Salvinia* – Prothallus

(3) Viroids – RNA

(4) Mustard – Synergids

38. Gymnosperms are also called soft wood spermatophytes because they lack **[AIPMT (Prelims)-2012]**

(1) Thick-walled tracheids

(2) Xylem fibres

(3) Cambium

(4) Phloem fibres

39. Which one of the following is common to multicellular fungi, filamentous algae and protonema of mosses? **[AIPMT (Prelims)-2012]**

(1) Multiplication by fragmentation

(2) Diplontic life cycle

(3) Members of kingdom Plantae

(4) Mode of Nutrition

40. *Cycas* and *Adiantum* resemble each other in having: **[AIPMT (Prelims)-2012]**

(1) Cambium (2) Vessels

(3) Seeds (4) Motile Sperms

41. Which one of the following is a **correct** statement? **[AIPMT (Prelims)-2012]**

(1) Antheridiophores and archegoniophores are present in pteridophytes

(2) Origin of seed habit can be traced in pteridophytes

(3) Pteridophyte gametophyte has a protonemal and leafy stage

(4) In gymnosperms, female gametophyte is free-living

42. Read the following five statements (A-E) and answer as asked next to them

(A) In *Equisetum*, the female gametophyte is retained on the parent sporophyte

(B) In *Ginkgo*, male gametophyte is not independent

(C) The sporophyte in *Riccia* is more developed than that in *Polytrichum*

(D) Sexual reproduction in *Volvox* is isogamous

(E) The spores of slime molds lack cell walls

How many of the above statements are **correct**? **[AIPMT (Mains)-2012]**

(1) Two (2) Three

(3) Four (4) One

43. *Selaginella* and *Salvinia* are considered to represent a significant step toward evolution of seed habit because **[AIPMT (Mains)-2011]**

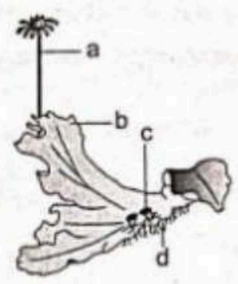
(1) Megaspores possess endosperm and embryo surrounded by seed coat

(2) Embryo develops in female gametophyte which is retained on parent sporophyte

(3) Female gametophyte is free and gets dispersed like seeds

(4) Female gametophyte lacks archegonia

44. Examine the figure given below and select the right option giving all the four parts (a, b, c, d) correctly identified **[AIPMT (Mains)-2011]**



(1) a – Seta
b – Sporophyte
c – Protonema
d – Rhizoids

(2) a – Antheridiophore
b – Male thallus
c – Globule
d – Roots

(3) a – Archegoniophore
b – Female thallus
c – Gemmacup
d – Rhizoids

(4) a – Archegoniophore
b – Female thallus
c – Bud
d – Foot

45. Consider the following four statements whether they are correct or wrong?

(A) The sporophyte in liverworts is more elaborate than that in mosses

(B) *Salvinia* is heterosporous

(C) The life-cycle in all seed bearing plants is diplontic

(D) In *Pinus*, male and female cones are borne on different trees

The two wrong statements together are **[AIPMT (Mains)-2011]**

(1) (B) and (C)

(2) (A) and (B)

(3) (A) and (C)

(4) (A) and (D)

46. Archegoniophore is present in : **[AIPMT (Prelims)-2011]**

(1) *Funaria*

(2) *Marchantia*

(3) *Chara*

(4) *Adiantum*

47. The gametophyte is **not** an independent, free - living generation in: **[AIPMT (Prelims)-2011]**

(1) *Pinus*

(2) *Polytrichum*

(3) *Adiantum*

(4) *Marchantia*

48. Compared with the gametophytes of the bryophytes, the gametophytes of vascular plants tend to be **[AIPMT (Prelims)-2011]**

(1) Smaller and to have smaller sex organs

(2) Smaller but to have larger sex organs

(3) Larger but to have smaller sex organs

(4) Larger and to have larger sex organs

49. Which one of the following is monoecious ? **[AIPMT (Mains)-2010]**

(1) *Marchantia*

(2) *Cycas*

(3) *Pinus*

(4) Date palm

50. Male and female gametophytes are independent and free-living in **[AIPMT (Prelims)-2010]**

(1) *Sphagnum* (2) Mustard

(3) Castor (4) *Pinus*

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51. Examine the figures A, B, C and D. In which one of the four options all the items A, B, C and D are correct? **[AIPMT (Mains)-2010]**

A. 

B. 

C. 

D. 

	A	B	C	D
(1)	Chara	Marchantia	Fucus	Pinus
(2)	Equisetum	Ginkgo	Selaginella	Lycopodium
(3)	Selaginella	Equisetum	Salvinia	Ginkgo
(4)	Funaria	Adiantum	Salvinia	Riccia

52. Which one of the following plants is monoecious? **[AIPMT (Prelims)-2009]**

(1) *Pinus* (2) *Cycas*
(3) Papaya (4) *Marchantia*

53. Which one of the following has haplontic life cycle? **[AIPMT (Prelims)-2009]**

(1) *Polytrichum* (2) *Ustilago*
(3) Wheat (4) *Funaria*

54. Phylogenetic system of classification is based on: **[AIPMT (Prelims)-2009]**

(1) Morphological features
(2) Chemical constituents
(3) Floral characters
(4) Evolutionary relationships

55. Mannitol is the stored food in: **[AIPMT (Prelims)-2009]**

(1) *Porphyra*
(2) *Fucus*
(3) *Gracillaria*
(4) *Chara*

56. Which one of the following is considered important in the development of seed habit? **[AIPMT (Prelims)-2008]**

(1) Heterospory
(2) Haplontic life cycle
(3) Free-living gametophyte
(4) Dependent sporophyte

57. Select one of the following of important features distinguishing *Gnetum* from *Cycas* and *Pinus* and showing affinities with angiosperms **[AIPMT (Prelims)-2008]**

(1) Embryo development and apical meristem
(2) Absence of resin duct and leaf venation
(3) Presence of vessel elements and absence of archegonia
(4) Perianth and two integuments

58. Which one of the following is heterosporous? **[AIPMT (Prelims)-2008]**

(1) *Equisetum*
(2) *Dryopteris*
(3) *Salvinia*
(4) *Adiantum*

59. In which one of the following male and female gametophytes do not have free living independent existence? **[AIPMT (Prelims)-2008]**

(1) *Cedrus* (2) *Pteris*
(3) *Funaria* (4) *Polytrichum*

60. In the prothallus of vascular cryptogam, the antherozoids and eggs mature at different times. As a result: **[AIPMT (Prelims-2007]**

(1) Self fertilization is prevented
(2) There is no change in success rate of fertilization
(3) There is high degree of sterility
(4) One can conclude that the plant is apomictic

61. If you are asked to classify the various algae into distinct groups, which of the following characters you should choose? **[AIPMT (Prelims)-2007]**

(1) Chemical composition of the cell wall
(2) Types of pigments present in the cell
(3) Nature of stored food materials in the cell
(4) Structural organization of thallus

62. Flagellated male gametes are present in all the three of which one of the following sets? **[AIPMT (Prelims)-2007]**

(1) *Riccia*, *Dryopteris* and *Cycas*
(2) *Anthoceros*, *Funaria* and *Spirogyra*
(3) *Zygnema*, *Saprolegnia* and *Hydrilla*
(4) *Fucus*, *Marsilea* and *Calotropis*

63. In gymnosperms, the pollen chamber represents: **[AIPMT (Prelims)-2007]**

(1) The microsporangium in which pollen grains develop
(2) A cell in the pollen grain in which the sperms formed
(3) A cavity in the ovule in which pollen grains are stored after pollination
(4) An opening in the mega gametophyte through which the pollen tube approaches the egg

64. Spore dissemination in some liverworts is aided by: **[AIPMT (Prelims)-2007]**

(1) Peristome teeth (2) Elaters
(3) Indusium (4) Calyptra

65. Moss peat is used as a packing material for sending flowers and live plants to distant places because : **[AIPMT (Prelims)-2006]**

(1) It is easily available
(2) It is hygroscopic
(3) It reduces transpiration
(4) It serves as a disinfectant

66. Conifers differ from grasses in the : **[AIPMT (Prelims)-2006]**

(1) Production of seeds from ovules
(2) Lack of xylem tracheids
(3) Absence of pollen tubes
(4) Formation of endosperm before fertilization

67. In a moss, the sporophyte **[AIPMT (Prelims)-2006]**

(1) Is partially parasitic on the gametophyte
(2) Produces gametes that give rise to the gametophyte
(3) Arises from a spore produced from the gametophyte
(4) Manufactures food for itself, as well as for the gametophyte

68. Match items in **column-I** with those in **column-II**

Column-I	Column-II
a. Peritrichous flagellation	(i) *Ginkgo*
b. Living fossil	(ii) *Macrocystes*
c. Rhizophore	(iii) *Escherichia coli*
d. Smallest flowering plant	(iv) *Selaginella*
e. Largest perennial alga	(v) *Wolffia*

Select the **correct** answer from the following : **[AIPMT (Prelims)-2005]**

(1) a(ii),b(i), c(iii), d(iv), e(v)
(2) a(v), b(iii), c(ii), d(v), e(i)
(3) a(i), b(ii), c(v), d(iii), e(ii)
(4) a(iii), b(i), c(iv), d(v), e(ii)

69. Top-shaped multiciliate male gametes and the mature seed which bears only one embryo with two cotyledons, are characteristic features of : **[AIPMT (Prelims)-2005]**

(1) Polypetalous angiosperms
(2) Gamopetalous angiosperms
(3) Conifers
(4) Cycads

Questions asked prior to Medical Ent. Exams. 2005

70. A system of classification, in which a large number of traits are considered, is

(1) Natural system
(2) Phylogenetic system
(3) Artificial system
(4) Synthetic system

71. Phenetic classification is based on

(1) The ancestral lineage of existing organisms
(2) Observable characteristics of existing organisms
(3) Dendrograms based on DNA characteristics
(4) Sexual characteristics

72. Which one of the following is wrongly matched?

(1) *Nostoc*-Water blooms
(2) *Spirogyra*-Motile gametes
(3) *Sargassum*-Chlorophyll *c*
(4) Basidiomycetes-Puffballs

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73. In chlorophyceae, the mode of sexual reproduction is
(1) Isogamy (2) Anisogamy
(3) Oogamy (4) All of these

74. An alga as the source of protein is
(1) *Chlorella* (2) *Nostoc*
(3) *Spirogyra* (4) *Ulothrix*

75. *Ulothrix* can be described as a
(1) Filamentous alga lacking flagellated reproductive stages
(2) Coenobial alga producing zoospores
(3) Filamentous alga with flagellated reproductive stages
(4) Non-motile colonial alga lacking zoospores

76. Sexual reproduction in *Spirogyra* is an advanced feature because it shows
(1) Different sizes of motile sex organs
(2) Same size of motile sex organs
(3) Morphologically different sex organs
(4) Physiologically differentiated sex organs

77. The pyrenoids are made up of
(1) Proteinaceous centre and starchy sheath
(2) Core of nucleic acid surrounded by protein sheath
(3) Core of protein surrounded by fatty sheath
(4) Core of starch surrounded by sheath of protein

78. Brown algae is characterised by the presence of
(1) Fucoxanthin (2) Haematochrome
(3) Phycocyanin (4) Phycoerythrin

79. *Ulothrix* filaments produce
(1) Heterogametes (2) Basidiospores
(3) Isogametes (4) Anisogametes

80. The plant body is thalloid in
(1) *Funaria* (2) *Sphagnum*
(3) *Salvinia* (4) *Marchantia*

81. In bryophytes
(1) Both generations are independent
(2) Gametophytes are dependent upon sporophytes
(3) Sporophytes complete their life cycle
(4) Sporophytes are dependent upon gametophytes

82. Which of the following plant kingdom is called 'amphibians'?
(1) Gymnosperm (2) Thallophyta
(3) Tracheophyta (4) Bryophyta

83. Bryophytes can be separated from algae, because they
(1) Possess archegonia
(2) Contain chloroplast
(3) Are thalloid forms
(4) Have no conducting tissue

84. Bryophytes are dependent on water, because
(1) Water is essential for their vegetative propagation
(2) The sperms can easily reach upto egg in the archegonium
(3) Archegonium has to remain filled with water for fertilization
(4) Water is essential for fertilization for their homosporous nature

85. Bryophytes comprise
(1) Dominant phase of gametophyte which produces spores
(2) Small sporophyte phase and generally parasitic on gametophyte
(3) Sporophyte is of longer duration
(4) Dominant phase of sporophyte which is parasitic

86. The antherozoids of *Funaria* are
(1) Multiciliated (2) Monociliated
(3) Aciliated (4) Biciliated

87. The plant body of moss *(Funaria)* is
(1) Completely sporophyte
(2) Predominantly gametophyte with sporophyte
(3) Completely gametophyte
(4) Predominantly sporophyte with gametophyte

88. Which of the following is true about bryophytes?
(1) They are thalloid
(2) They possess archegonia
(3) They contain chloroplast
(4) All of these

89. Multicellular branched rhizoids and leafy gametophytes are the characteristics of
(1) Some bryophytes
(2) Pteridophytes
(3) All bryophytes
(4) Gymnosperms

90. Elater for spore dispersal mechanism is exhibited by
(1) Liverworts
(2) *Marchantia*
(3) *Riccia*
(4) *Funaria*

91. The walking fern is so named because
(1) It propagates vegetatively by its leaf tips
(2) It knows how to walk by itself
(3) Its spores are able to walk
(4) It is dispersed through the agency of walking animals

92. In which of the following would you place the plants having vascular tissue lacking seeds?
(1) Pteridophytes
(2) Gymnosperms
(3) Algae
(4) Bryophytes

93. Which aquatic fern performs nitrogen fixation?
(1) *Azolla*
(2) *Nostoc*
(3) *Salvia*
(4) *Salvinia*

94. Plants reproducing by spores such as mosses and ferns are grouped under the general term
(1) Cryptogams
(2) Bryophytes
(3) Sporophytes
(4) Thallophytes

95. Which one of the following is a vascular cryptogam?
(1) *Cedrus*
(2) *Equisetum*
(3) *Ginkgo*
(4) *Marchantia*

96. Heterospory and seed habit are often exhibited by a plant possessing
(1) Petiole
(2) Ligule
(3) Bract
(4) Spathe

97. What is common in all the three, *Funaria*, *Dryopteris* and *Ginkgo*?
(1) Independent sporophyte
(2) Presence of archegonia
(3) Well developed vascular tissues
(4) Independent gametophyte

98. In *Pinus*, the wing of the seed develops from
(1) Ovuliferous scale
(2) Integument
(3) Nucellus
(4) Bract

99. Which one of the following statements about *Cycas* is incorrect?
(1) It has circinate vernation
(2) Its xylem is mainly composed of xylem vessel
(3) Its roots contain blue-green algae
(4) It does not have a well organized female flower

100. Largest sperms in the plants world are found in
(1) Banyan
(2) *Cycas*
(3) *Thuja*
(4) *Pinus*

101. Transfusion tissue is present in the leaves of
(1) *Pinus*
(2) *Dryopteris*
(3) *Cycas*
(4) Both (1) & (3)

102. The endosperm of gymnosperm is
(1) Diploid
(2) Polyploid
(3) Triploid
(4) Haploid

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103. Plant group with largest ovule, largest tree, and largest gametes is
(1) Gymnosperm (2) Angiosperm
(3) Bryophyta (4) Pteridophyta

104. Which of the following plants produce seeds but not flowers?
(1) Maize (2) Mint
(3) Peepal (4) *Pinus*

105. *Cycas* have two cotyledons but not included in angiosperms because of
(1) Naked ovules (2) Seeds like monocot
(3) Circinate ptyxis (4) Compound leaves

106. Which one of the following is a living fossil?
(1) *Cycas*
(2) Moss
(3) *Saccharomyces*
(4) *Spirogyra*

107. Which of the following is without exception in angiosperms?
(1) Presence of vessels (2) Syngamy
(3) Secondary growth (4) Autotrophic nutrition

108. Which one of the following pairs of plants are not seed producers ?
(1) Fern and *Funaria*
(2) *Funaria* and *Ficus*
(3) *Ficus* and *Chlamydomonas*
(4) *Punica* and *Pinus*

109. Angiosperms have dominated the land flora primarily because of their
(1) Power of adaptability in diverse habitat
(2) Property of producing large number of seeds
(3) Nature of self pollination
(4) Domestication by man

110. Transport of food material in higher plants takes place through
(1) Companion cells (2) Transfusion tissue
(3) Tracheids (4) Sieve elements

111. An alga which can be employed as food for human being is
(1) *Ulothrix*
(2) *Chlorella*
(3) *Spirogyra*
(4) *Polysiphonia*

SECTION - D

NEET Booster Questions

1. Find out the **correct** sequence (w.r.t. size of main gametophyte in decreasing manner)
(1) *Zea mays*, *Eucalyptus*, *Pteris*
(2) *Selaginella*, *Pinus*, *Mangifera*
(3) *Marchantia*, *Triticum*, *Spirogyra*
(4) *Dryopteris*, *Sphagnum*, *Ulothrix*

2. Consider the following structures/characters:
(i) Multicellular asexual buds
(ii) Protonema stage
(iii) Sporic meiosis
(iv) Multiflagellate sperms
(v) Vascular amphibians of plant kingdom
The structures/characters concerned with the life cycle of liverworts are:
(1) (i), (iii) and (v) (2) (iii) & (v)
(3) (ii), (iv) & (v) (4) (i) & (iii)

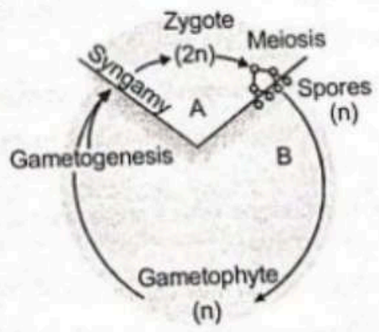
3. Beginning with germination of spores in life cycle of *Dryopteris*, what is the sequence of structures that develop after germination?
(a) Spore mother cell (b) Sporophyte plant
(c) Gametes (d) Gametophyte
(e) Embryo (f) Zygote
(1) (a), (c), (f), (e) & (b)
(2) (a), (d), (c), (f), (e) & (b)
(3) (d), (c), (f), (b), (e) & (a)
(4) (d), (c), (f), (e), (b) & (a)

4. The spread of living pteridophytes is limited and restricted to narrow geographical regions because
(i) Gametophyte is inconspicuous, multicellular and free-living
(ii) They have delicate and graceful foliage structure

(iii) Sperms require an external supply of water for swimming and reaching the non-jacketed female sex organ
(iv) Gametophytes require cool, damp and shady places to grow
Mark the correct option(s)
(1) (iii) & (iv)
(2) (i), (iii) & (iv)
(3) (ii) & (iv)
(4) Only (iv)

5. *Fucus* and *Volvox* resemble each other in presence of
(1) Non-sulphated phycocolloids in cell wall
(2) Internal fertilisation and gametic meiosis
(3) Large and non-motile female gamete in oogonium
(4) Air bladders in the region of branching

6. See the given life-cycle pattern and choose the option which correctly represent the characters for the organism.



(1) *Chara* – Zygotic meiosis, free living sporophyte
(2) *Chlamydomonas* – Haplontic lifecycle, sporic meiosis
(3) *Spirogyra* – Haplontic lifecycle, no multicelled sporophyte
(4) *Ulothrix* – Gametic meiosis, diplontic lifecycle

7. Go through the following matches
(a) *Spirullina* – Rich source of protein
(b) *Gelidium* – Polysulphate esters in cell wall
(c) *Chara* – Unicellular green alga
(d) *Polysiphonia* – Haplontic life cycle
Which of these are correct?
(1) (b) & (c)
(2) (a) & (b)
(3) (c) & (d)
(4) (a), (b) & (c)

8. In which of the following set of plants, development of 7 celled and 8 nucleated gametophyte is preceded by reduction division as well as mitosis?
(1) *Pteris*, *Adiantum*
(2) *Pisum*, *Eucalyptus*
(3) *Chlamydomonas*, *Dryopteris*
(4) *Azolla*, *Ficus*

9. Branch of taxonomy in which DNA sequences, alkaloids, crystals may help in knowing relationships among organisms is
(1) Karyotaxonomy (2) Chemotaxonomy
(3) Cytotaxonomy (4) Morphotaxonomy

10. All statements are true for both ferns and conifers, **except**
(1) True stem, leaves and roots in main plant body
(2) Independent free living existence of gametophytes
(3) Archegoniate embryophytes
(4) Oogamous sexual reproduction

11. How many of the following feature(s) is/are commonly concerned with *Pinus* as well as Cycas?
(a) Pinnate leaves (b) Mycorrhiza
(c) Branched stem (d) Wind pollination
(e) Female strobili (f) Non-motile sperms
(g) Simple leaves (h) Naked ovule
Mark the **correct** option.
(1) Four (2) Three
(3) Two (4) One

12. Ferns and mosses resemble in possessing
(1) Large sized true leaves
(2) Unicellular rhizoids
(3) Haplo-diplontic life cycle pattern
(4) Gametic meiosis in lifecycle

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13. Which of the following is a **wrong** combination?
    (1) Cell wall is made up of cellulose and algin – Brown algae
    (2) Strobilus, haplontic life cycle, biflagellate sperm – Horsetails
    (3) Non motile gametes – Red algae
    (4) Coralloid roots, dioecious – Cycadales plants

14. What will be the number of chromosomes in endosperm (p) and perisperm (q) if there are 12 chromosomes in venter canal cell of a conifer?
    (1) (p) 36, (q) 24 (2) (p) 24, (q) 24
    (3) (p) 36, (q) 12 (4) (p) 12, (q) 24

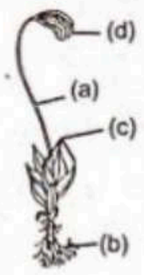
15. In which of the following plant(s), dominant phase is the independent, photosynthetic, thalloid, dorsiventral haploid gametophyte which alternates with the short-lived multicelled sporophyte totally dependent on gametophyte?
    (1) *Funaria*, *Sphagnum* (2) *Marchantia*, *Riccia*
    (3) *Pteris*, *Porella* (4) Both (1) & (2)

16. Heterosporous vascular cryptogams fail to develop seeds because
    (1) Female gametes are large and non-motile
    (2) Retention of megaspores permanently within the megasporangia has not become established
    (3) Fusion of gametes results in formation of zygote which develops into embryo
    (4) They lack precocious germination of spores

17. Which of the following structure/feature is common in *Volvox*, mosses, ferns and conifers?
    (1) True leaves
    (2) Vascular bundles
    (3) Embryo
    (4) Oogamous reproduction

18. *Ginkgo* and *Cedrus* are included in gymnosperms because
    (1) Ovary is not enclosed by any wall and remains exposed, both before and after fertilisation
    (2) Seeds are not protected by seed coats
    (3) Seeds do not occur inside the fruit
    (4) Both (2) and (3)

19. Given picture is of a bryophyte. The correct ploidy levels of the indicated structures are



    (1) (a) : 2n, (b) : 2n, (c) : n, (d) : n
    (2) (a) : n, (b) : n, (c) : n, (d) : 2n
    (3) (a) : 2n, (b) : n, (c) : n, (d) : 2n
    (4) (a) : 2n, (b) : n, (c) : 2n, (d) : 2n

20. If a root cell of a pteridophyte contains 10 chromosomes, then the chromosome number in its leaf cell, sperm mother cell and spore mother cell respectively is
    (1) 10, 10, 10
    (2) 5, 10, 10
    (3) 10, 5, 10
    (4) 10, 5, 5

21. Vascular phanerogams with naked seeds have
    (1) Haploid phase which is independent and free living
    (2) Reproductive leaves aggregated to form cones or strobilus
    (3) Pyrenoids in their chloroplast
    (4) Glycogen as stored food

22. *Selaginella* and *Salvinia* have/are
    (1) Unisexual gametophytes
    (2) Produce only one types of spores
    (3) Unicellular thalloid gametophyte
    (4) Terrestrial heterosporous ferns

23. Fertilisation is siphonogamous with non-motile male gametes in
    (1) *Adiantum* and *Gnetum*
    (2) *Ficus* and *Fucus*
    (3) *Cycas* and *Dryopteris*
    (4) *Pinus* and *Calotropis*

24. Sexual reproduction is oogamous in
    (1) *Volvox* and *Fucus*
    (2) *Spirogyra* and *Ulothrix*
    (3) *Polysiphonia* and *Spirogyra*
    (4) *Ulothrix* and *Chara*

25. Pollination in gymnosperm is
    (1) Anemophilous and indirect
    (2) Hydrophilous and direct
    (3) Anemophilous and direct
    (4) Entomophilous and direct

26. Sexual system of classification is
    (1) Artificial system
    (2) Based on stamens characters
    (3) Based on corolla and carpels characters
    (4) Both (1) & (2)

27. The Bentham and Hooker's classification is
    (1) Classification of taxa based on actual examination
    (2) Artificial system of classification
    (3) Phylogenetic system of classification
    (4) Based on evolution

28. The thallus organisation of *Volvox* is
    (1) Multicellular and coccoid
    (2) Colonial and nonflagellate
    (3) Unicellular
    (4) Colonial and motile

29. Brown algae are quite common in
    (1) Fresh water habitats
    (2) Tropical sea water
    (3) Temperate sea water
    (4) Both (2) & (3)

30. Algae with floridean starch as reserve food material is also characterised by
    (1) Presence of chlorophyll b
    (2) Stacked thylakoids
    (3) Nonsulphated phycocolloids
    (4) Nonflagellate nature

31. 100 zygospores, alternate with empty cells in *Spirogyra* are under _____ conjugation and the total number of daughter filaments formed will be _____
    (1) Scalariform, 400
    (2) Lateral, 100
    (3) Lateral, 400
    (4) Scalariform, 100

32. Algin is a phycocolloid, obtained from the cell wall of
    (1) *Macrocystis* and *Porphyridium*
    (2) *Mastigocladus* and *Laminaria*
    (3) *Microcystis* and *Nereocystis*
    (4) *Macrocystis* and *Fucus*

33. Which of the following is a red alga that is not red?
    (1) *Nemalion*
    (2) *Polysiphonia*
    (3) *Gelidium*
    (4) *Batrachospermum*

34. In chlorophyceae, the flagella are
    (1) Tinsel type
    (2) Whiplash type
    (3) Whiplash and tinsel type
    (4) Basal tinsel, apical whiplash type

35. Which of the following are useful for curing goitre?
    (1) Sea kelps
    (2) Diatoms
    (3) Red algae
    (4) *Porphyra*

36. Non-motile gametes are characteristically found in
    (1) Cyanophyta (2) Rhodophyta
    (3) Phaeophyta (4) Chlorophyta

37. The female sex organ in red algae is flask shaped and is known as
    (1) Trichogyne (2) Carpogonium
    (3) Spermatium (4) Archegonium

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38. Some characters of algae are given below

a. Floridean starch
b. Sulphated phycocolloids in cell wall
c. Alginic acid
d. Trumpet hypha
e. Haplodiplontic life cycle
f. Isomorphic alternation of generation
g. Fucoxanthin
h. Phycoerythrin
i. Zygotic meiosis
j. Two anterior flagella

Which of the given set of characters belongs to *Laminaria*?

(1) a, b, e, f, h (2) c, d, e, g
(3) b, c, d, e, f, g, i (4) c, d, e, f, g, i

39. Bryophytes are not characterised by

(1) Sporophyte parasitic over gametophyte
(2) Independent gametophyte
(3) Absence of vascular tissues
(4) Independent sporophyte

40. Stems and leaves of bryophytes are

(1) Analogous to vascular plants
(2) Homologous to vascular plants
(3) Analogous to algae & fungal thallus
(4) None of these

41. Non-vascular embryophyte with leaves is

(1) *Riccia*
(2) *Porella*
(3) *Selaginella*
(4) *Macrocystis*

42. Find set of features related to *Funaria*

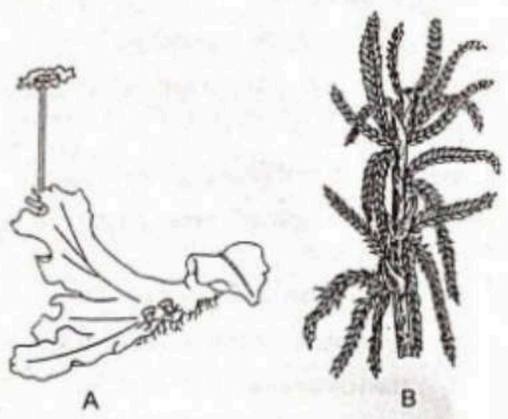
a. Protonema
b. Prothallus
c. Gametophore
d. Thallus body
e. NCC in antheridium
f. Haplodiplontic life cycle
g. True plant organs in sporophyte
h. Fragmentation

(1) b, d, e, g
(2) a, d, f, g
(3) a, c, f, g, h
(4) a, c, f, h

43. In *Funaria*, 20 chromosomes are present in rhizoids, then the number of chromosome in calyptra, theca and foot will be

(1) 20, 40, 40 respectively
(2) 40, 20, 20 respectively
(3) 20, 40, 20 respectively
(4) 40, 10, 20 respectively

44. Identify the plants A and B in the figures given below



(1) A - Female *Marchantia*, B - *Sphagnum*
(2) A - *Riccia*, B - *Marchantia*
(3) A - *Marchantia*, B - *Funaria*
(4) A - Male *Marchantia*, B - *Sphagnum*

45. Algae, bryophytes and pteridophytes resemble with each other in which one of the following feature?

(1) Gametophytic plant body
(2) Dependence on water for fertilisation
(3) Haplontic alternation of generation
(4) Presence of embryo

46. Find correct statement for the prothallus of fern

(1) Monoecious, protandrous with multicellular rhizoides
(2) Monoecious, protandrous with unicellular rhizoides
(3) Dioecious with unicellular rhizoides
(4) Monoecious, protandrous with apical antheredia and basal archegonia on ventral surface

47. The dominant photosynthetic phase in the life cycle of pteridophyta is equivalent to the

(1) Gametophytic phase of bryophyta
(2) Sporophytic phase of bryophyta
(3) Gametophytic phase of pteridophytes
(4) Gametophytic phase of gymnosperm

48. In Pteridophytes, reduction division occurs when

(1) Prothallus is formed
(2) Sex organs are formed
(3) Spores are formed
(4) Gametes are formed

49. The evolutionary advanced features of *Selaginella* are

(a) Heterospory
(b) Endosporic development of gametophyte
(c) Reduced gametophyte
(d) Localization of sporangium bearing appendages in strobili
(e) Unisexual gametophytes
(f) Fertilization with the help of water

(1) All are correct
(2) All except (f) is correct
(3) All except (e) and (f) are correct
(4) All except (c) is correct

50. How many structures listed below are diploid for a typical fern member?

a. Indusium cell b. Stomium cell
c. NCC d. Rhizome cell
e. Sporophyll cell f. Prothallus cell
g. SMC h. Spore
i. Antherozoid mother cell

(1) Nine (2) Six
(3) Five (4) Seven

51. Mark the correct statement for the organism given below in figure.



(1) The structure labelled A is male cone
(2) It is member of sphenopsida
(3) Nodes are hollow while internodes are solid
(4) This is commonly called as stonewort

52. Member of plantae having endospermic, perispermic, polycotyledonous and winged seeds is also related to

(1) Sulphur shower
(2) Largest ovule
(3) Double fertilization
(4) Placentation

53. Which one constitutes the dominant vegetation in colder regions?

(1) Monocots
(2) Dicots
(3) Legumes
(4) Gymnosperms

54. Which of the given sets are matched correctly?

a. *Chondrus* - Algin
b. *Gracilaria* - Agar
c. *Cycas* - Coralloid root
d. *Pinus* - Canada balsum
e. *Adiantum* - Walking fern
f. *Lycopodium* - Cord moss
g. *Cedrus* - Independent gametophyte
h. *Sequoia* - Tallest gymnosperm

(1) b, c, e, h (2) a, b, c, e, f
(3) b, c, e, g, h (4) b, c, d, e, g, h

55. How many generations are present in the seed of gymnosperm?

(1) 2 (2) 3
(3) 1 (4) 4

56. Gametophytic plant body is nonvascular in

(1) Algae and liverworts
(2) Mosses and ferns
(3) Gymnosperms and angiosperms
(4) All of these

57. Endosperm of gymnosperm is ontogenetically similar to angiospermic

(1) Endosperm (2) Embryo sac
(3) Archegonium (4) Megasporangia

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58. Flowering plants are more successful than other members of the plant world because

(1) They are large and have a good vascular tissue system

(2) They carry out variety of pollination mechanism

(3) The protected plant embryo can survive in the period of unfavourable conditions

(4) All of these

59. A. Heterospory is found in all members of pteropsida

B. *Selaginella* is advance among pteridophytes as it produces seeds

C. *Pinus* leaves are monomorphic, pinnate compound and have sunken stomata as adaptation against transpiration

D. Sporic meiosis is characteristic of life cycle in many organisms like *Volvox*, *Chlamydomonas* and *Ulothrix*.

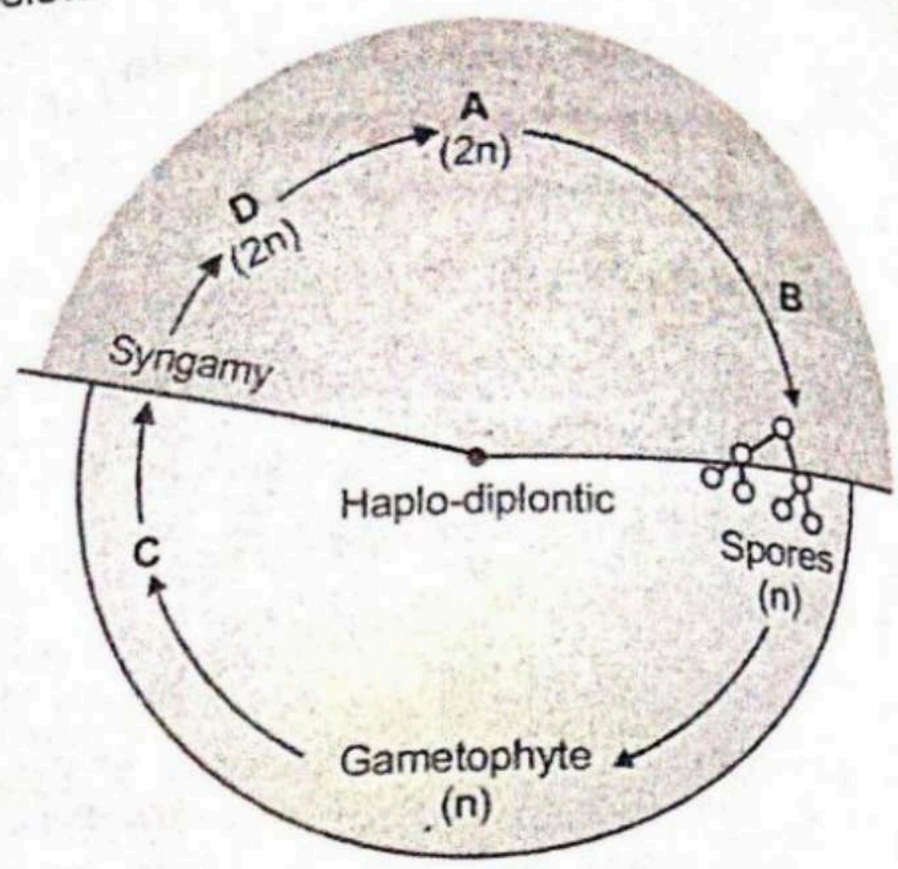
(1) All are incorrect

(2) Both B & C are correct

(3) Only B is correct

(4) Only D is incorrect

60. Identify the labels A, B, C and D in the figure given below



(1) A - Sporophyte; B - Meiosis
C - Gametogenesis; D - Endosperm

(2) A - Sporophyte; B - Mitosis
C - Gametogenesis; D - Zygote

(3) A - Gametophyte; B - Meiosis
C - Gametogenesis; D - Zygote

(4) A - Sporophyte; B - Meiosis
C - Gametogenesis; D - Zygote

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